Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kwale County.

Final Vanga Ward Adaptation Plan

Consultant

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1 Chapter 1: Background information

1.1 Introduction

Plan international Kenya is implementing the Conservation and Sustainable Management of Marine and Coastal Ecosystems (COSME) Project in Kwale and Kilifi Counties. The two counties are endowed with rich coastal and marine ecosystems ranging from terrestrial forests, mangroves, seagrass beds, coral reefs, marine species and open sea marine resources among others. These natural assets not only underpin livelihoods but also drive key economic sectors including tourism, agriculture, fisheries, and maritime trade. Despite their economic and ecological value, the 2021 country and county climate risk profiles highlight that these resources are increasingly under threat from climate change impacts such as drought, flooding, sea level rise, ocean acidification, rising temperatures, and forest fires. The consequences of these risks exacerbate underlying socio-economic vulnerabilities, with women, youth, and Persons Abled Differently (PAD) experiencing disproportionate impacts due to structural inequalities. Loss of income, displacement, reduced productivity, and damage to infrastructure and markets are already affecting community well-being and resilience in the region.

In response, the COSME project aims to enhance women-led, climate-resilient livelihoods through ecosystem-based approaches. Hence, to support project implementation requires the development of Gender Responsive Adaptation and Preparedness Plans (GRAPPs) that integrate climate risk diagnostics with locally appropriate adaptation pathways. The Vanga Ward Climate Change Adaptation Plan (WAP) is therefore prepared as part of the GRAPPs outputs. The ward which measures 114.9 km^2 Vanga Ward, is located in Kwale County at Kenya's southern coast with a population of 24,616 (12,189 male & 12,425 female) (KNBS, 2019) (Figure 1), faces a high vulnerability to climate change impacts. The ward's The climate is a tropical coastal characterized by bimodal rainfall and warm temperatures year-round. It's economy and livelihoods depend heavily on climate-sensitive sectors such as rain-fed agriculture, livestock, and coastal resources (Figure 1). However, recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode development gains and degrade critical natural resources upon which local livelihoods are dependent, making adaptation a critical priority for the ward.

Recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode hard-won development gains, making adaptation a critical priority for the ward. In response, Kenya's policy framework has evolved to support local climate action, from the National Climate Change Response Strategy and Action Plans to the Climate Change Act 2023, which emphasizes devolved climate governance and resilience building at county and community levels. This Ward Adaptation Plan aligns with national and county climate change policies by mainstreaming adaptation into local planning and empowering the community to implement priority resilience measures.

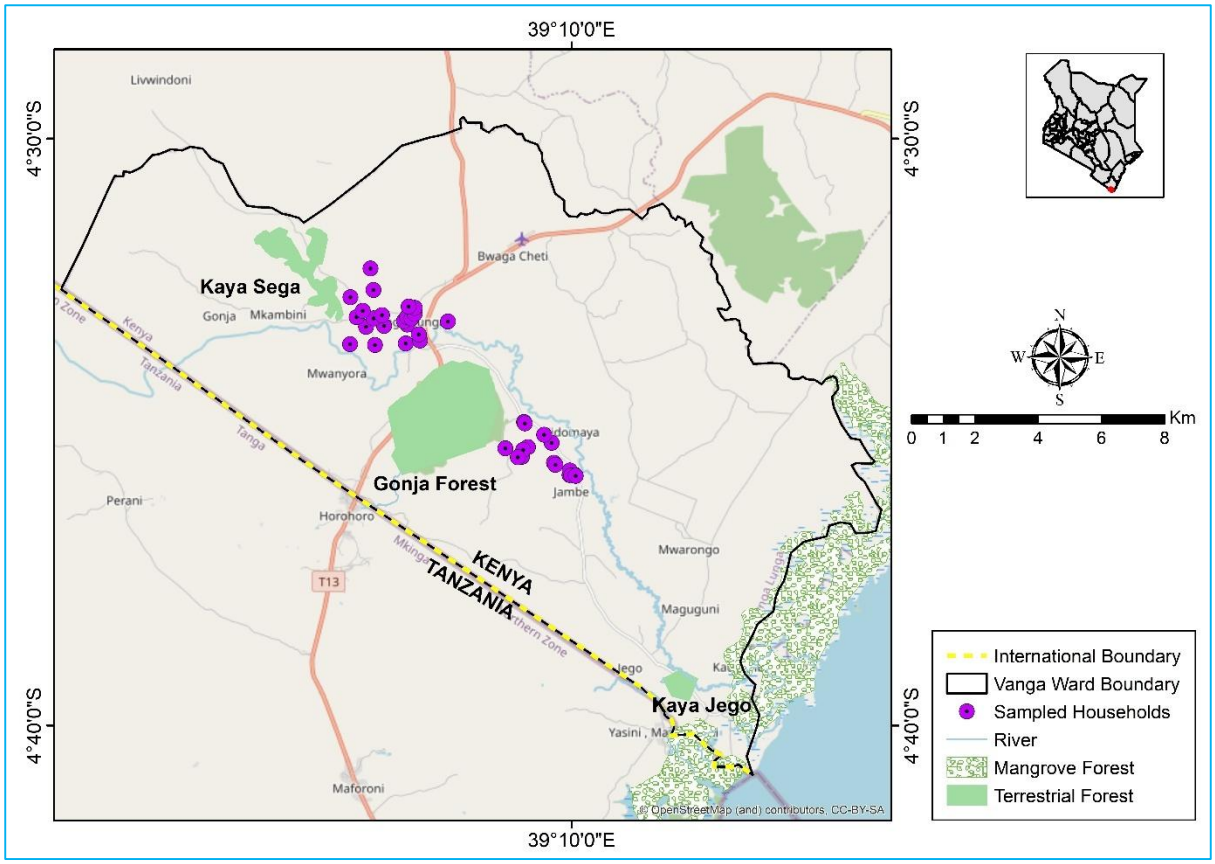

Figure 1:Location of Vanga Ward and survey sites (Source; PCVA, 2025)

1.2 Plan development approach

The Ward Adaptation Plan for Vanga was developed in a participatory, gender-responsive and livelihoods-based vulnerability framework that conceptualized climate risk as an outcome of the interaction between exposure, sensitivity and adaptive capacity (Figure 2). Within this analytical framing, vulnerability was understood as the degree of climate-induced damage experienced by socio-ecological systems, while resilience denoted the system's ability to maintain or rapidly regain functionality following climatic shocks. This approach acknowledged that climate risks in Vanga, characterized predominantly by recurrent droughts, seasonal flooding, strong winds, sea-level rise, pests and diseases and human–wildlife conflicts arising from proximity to River Umba, do not affect population groups uniformly. Women, men, youth, elderly persons and Persons Abled Differently (PADs) face differentiated risks due to disparities in livelihood assets, labor burdens, mobility, access to resources, intra-household power dynamics and decision-making autonomy.

The plan was therefore designed using a process that combined experiential community knowledge with quantitative datasets, spatial analysis, ecological diagnostics and climatic modelling. This integrated methodology enabled the plan to link climatic hazards to specific livelihood systems; rainfed agriculture, artisanal fishing, seaweed farming, mangrove harvesting, small-scale trade and casual labour and to examine how these systems are shaped by gender norms, social vulnerabilities and institutional arrangements at ward level. The resulting insights provided a nuanced understanding of how climate extremes intersect with poverty, food insecurity, ecosystem degradation, weak infrastructure and limited institutional support, thereby shaping the context within which adaptation strategies must be developed.

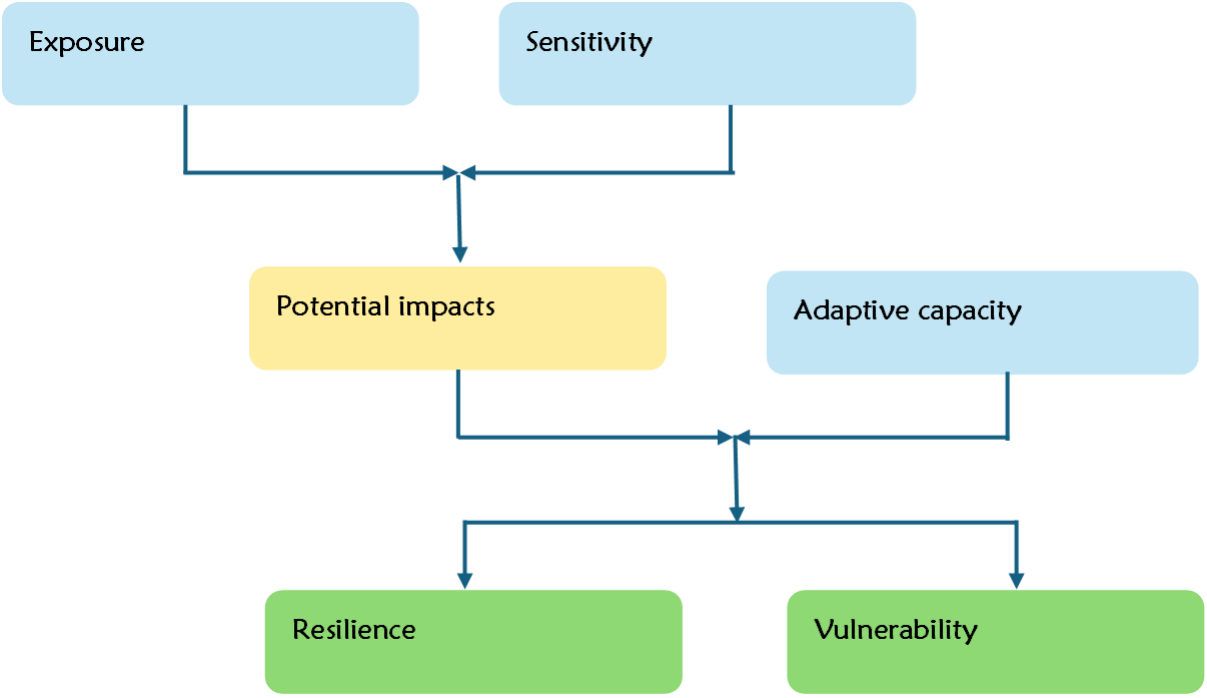

Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014).

1.2.1 Inception Phase

The methodology was agreed upon through an inception phase that established a shared understanding among Plan International Kenya and the consultant on the objectives, scope and analytical expectations of the PCVA and WAP processes. An inception meeting held on 5th September 2025 brought together the COSME project teams to refine the methodological sequence, expected results, deliverable structure and timelines. During this phase, existing climate risk information for Kwale County was reviewed, data gaps specific to Vanga Ward were identified, and a sequenced work plan was developed outlining participatory diagnostics, household surveys, spatial and climate analysis, and subsequent synthesis and validation activities.

Following institutional alignment, data collection instruments were developed and technically refined to ensure that they were methodologically sound, gender-responsive and culturally appropriate. The instruments included ward-level FGD guides segregated for women, men and youth; participatory mapping and historical timeline guides; seasonal calendar and daily clock templates; household decision-making pile-sorting tools; Leaky Bucket diagrams; impact chain protocols; vulnerability matrix templates; institutional Venn diagram formats; and adaptation pathway worksheets. Each tool integrated prompts on gender norms, intersectional vulnerability, unpaid care workloads, labour shifts during climate stress and disability inclusion.

Stakeholder mapping during the inception phase identified relevant actors within Vanga Ward, including the Ward Climate Change Planning Committee, chiefs and assistant chiefs, village elders, leaders of women and youth groups, PAD representatives and support networks, seaweed farming groups, mangrove user groups, BMUs, WRUAs and community-based organizations. These actors facilitated community mobilization, venue preparation and dissemination of study information at ward level. The inception phase ensured that ward stakeholders were adequately oriented and positioned to meaningfully participate in the PCVA and adaptation planning process.

1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA)

The Ward Adaptation Plan was informed by a comprehensive PCVA process implemented in four COSME target villages within Vanga Ward i.e., Mikoroshoni,Mwalewa C, Ngweneni and Sega D. The assessment followed a structured six-step participatory sequence designed to diagnose climate vulnerabilities, identify underlying drivers of sensitivity, assess existing coping and adaptation strategies, and translate diagnostic findings into actionable adaptation options suitable for inclusion in the WAP. The process applied the full suite of ten PCVA tools, each deployed to generate gender- and age-disaggregated evidence capable of capturing complex interactions between climate hazards, livelihoods, natural resources and social systems.

1.2.2.1 Step 1: Scoping

During scoping, the geographical scope of the ward was delineated using GIS shapefiles provided by Plan International and Kwale County Government. Within this boundary, the four COSME villages were purposively selected on the basis of ecological sensitivity, concentration of climate-sensitive livelihoods, recurrent exposure to droughts and floods, susceptibility to strong winds and sea-water intrusion, and documented human–wildlife conflict incidents, particularly crocodile attacks reported around River Umba profiles were then developed to summarize demographic characteristics, livelihood systems, existing infrastructure, ecosystem conditions, climate-related impacts, past interventions and vulnerability hotspots. These profiles served as the baseline data against which PCVA tools were deployed.

1.2.2.2 Step 2: Establishment of Core Participatory Analysis Teams

A ward-level core participatory analysis team was established for Vanga, drawing membership from Ward Climate Change Planning Committee (WCCPC) representatives, village elders, women leaders, youth leaders, PAD representatives and carers, NbS groups, GJJ groups, and other active CBOs and self-help groups. County technical officers from climate change, environment, forestry, agriculture, livestock and fisheries were engaged as resource persons.

The core team underwent a structured orientation on the PCVA framework, the ten PCVA tools, gender- and age-responsive facilitation, child safeguarding and ethical data collection. Training sessions used a combination of plenary presentations, group work and simulations of each PCVA tool, conducted primarily in Kiswahili to ensure comprehension. By the end of the training, the ward team had the capacity to co-facilitate hazard and resource mapping, historical timelines, seasonal calendars, daily clocks, household decision making and pile-sorting exercises, Leaky Bucket analyses, impact chain mapping, vulnerability matrix scoring, Venn diagrams and adaptation pathway development.

1.2.2.3 Step 3: Situation Analysis of Climate Vulnerability

The situation analysis integrated secondary datasets, primary field evidence and participatory analytics to generate a comprehensive picture of exposure, sensitivity and adaptive capacity across Vanga Ward. This phase synthesised literature review, ward-level gender- and age-disaggregated Focus Group Discussions (FGDs), transect walks across the five COSME villages, and the structured application of PCVA Tools 1–7. Through triangulation, these methods produced a detailed, spatially nuanced and socially differentiated understanding of how climatic hazards interact with biophysical systems, livelihood structures and gendered social dynamics.

1.2.2.3.1 Literature Review

The literature review served as the technical backbone of the situation analysis, drawing from global, regional, national, county and historical ward-level evidence to contextualize observed vulnerabilities. At the global scale, the review considered findings from IPCC AR6 Working Group I and II, particularly projections relevant to East African coastal systems, including

accelerated warming rates, sea surface temperature rise, intensification of short-duration extreme rainfall, increased occurrence of coastal storms, saline intrusion risks and anticipated sea-level rise along the Western Indian Ocean. These findings were aligned with global adaptation and risk governance frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030, the Paris Agreement adaptation cycle, and SDGs 2, 5, 6, 13 and 14, which collectively frame resilience building in coastal and marine socio-ecological systems.

At the regional scale, the review integrated IGAD Climate Prediction and Applications Centre (ICPAC) bulletins, the Africa Climate Policy Centre (ACPC) projections for Eastern Africa, and Western Indian Ocean Marine Science Association (WIOMSA) research on mangrove decline, seaweed vulnerability, fishery collapses and oceanographic shifts. This regional evidence positioned Vanga within a broader coastal climate risk corridor characterized by rising temperatures, altered monsoon circulation, changing upwelling systems and degradation of nearshore ecosystems.

National climate policy and regulatory frameworks were examined comprehensively, including the Climate Change Act (2023), National Adaptation Plan (NAP 2015–2030), Kenya's Updated NDC (2023), the National Framework for Climate Services (NFCS), Kenya Climate-Smart Agriculture Strategy, National Disaster Management Policy, National Drought Early Warning System, and EMCA (2015) Cap 387 of the Laws of Kenya and sectoral regulations. These documents provided the lens for interpreting institutional mandates, priority hazards and adaptation responsibilities that cascade to county and ward levels.

At county scale, the review consolidated information from the Kwale County Integrated Development Plan (CIDP III, 2023–2027), the Kwale County Climate Change Act (2022), the County Climate Change Fund (CCCF) Regulations, the County Climate Change Action Plan (CCCAP 2023–2027), the County Spatial Plan, and sector strategies in agriculture, livestock, fisheries, water and coastal management. These county frameworks were triangulated with socio-economic and food security trends reported by the National Drought Management Authority (NDMA), historical rainfall data from the Kenya Meteorological Department (KMD), marine ecosystem studies from KMFRI, forestry and mangrove assessments by the Kenya Forest Service (KFS), fishery catch records from the State Department for Fisheries, and demographic data from the Kenya Population and Housing Census (2019).

The review also incorporated previous PCVAs conducted under COSME and other coastal adaptation programmes, which provided comparative vulnerability patterns, gender-responsive insights, and past lessons on nature-based solutions, community-based early warning systems, and local adaptation practices. This multilevel synthesis created a scientifically grounded baseline against which primary data from Vanga could be interpreted.

1.2.2.3.2 Focus Group Discussions (FGDs)

Ward-level FGDs were conducted with segregated groups of women, men and youth from the four COSME villages;Mikoroshoni, Mwalewa C, Ngweneni and Sega D, to elicit gendered and generational insights into climate trends, livelihood impacts and socio-ecological vulnerability (Figure 3). The discussions explored perceived changes in rainfall reliability, including delayed onset and inconsistent cessation of rains, prolonged intra-seasonal dry spells, unprecedented periods of elevated temperatures, shifts in wind patterns and increased intensity of episodic storms and coastal surges. Participants described the material impacts of these climatic variations on agricultural productivity, seaweed quality and harvest cycles, artisanal fisheries (including reduced catch per unit effort), mangrove health, livestock mortality and food security.

The FGDs also highlighted the deeply gendered nature of climate vulnerability. Women described increased walking distances and time burdens for water and fuel collection during droughts; heightened caregiving demands during disease outbreaks; increased exposure to GBV during water scarcity periods; and disruptions in women's informal trading activities. Youth reported loss of wage labour opportunities during climate shocks, stress-related migration, and lack of access to climate-resilient livelihood assets. Men described reduced fishing days caused by strong winds, damaged boats during storms and increased anxiety over household provisioning responsibilities. In addition, participants highlighted PAD-specific vulnerabilities, including mobility limitations during floods and exclusion from labour-intensive coping strategies.

These discussions produced climate calendars, gender-disaggregated vulnerability matrices and hazard trend narratives that deepened the temporal and social dimensionality of the vulnerability assessment.

Figure 3: Focus Group Discussions in Vanga Ward (Source: Field Surveys,2025)

1.2.2.3.3 Transect Walks

Transect walks undertaken across the five villages provided on-ground spatial verification of vulnerabilities identified in FGDs and documented in literature. These walks traversed rainfed agricultural plots, irrigated horticultural fields, mangrove stands and adjacent intertidal zones, seaweed farming lines, fish landing sites, riparian zones of River Umba, boreholes and shallow wells, public institutions (schools, health dispensaries), access roads, market areas, and known flood plains.

Structured observation sheets and GPS-enabled devices were used to capture geospatial attributes of key vulnerability hotspots. Evidence recorded included deterioration of feeder roads due to repeated flooding, heavily eroded gullies formed by runoff from bare catchments, saline intrusion into agricultural fields and domestic wells, siltation of wetlands, collapsed seaweed lines following strong winds, degraded mangrove stands exhibiting canopy thinning or sediment stress, and infrastructure damage such as washed-out culverts and submerged pathways.

These observations enhanced the physical realism of the vulnerability analysis and provided geo-referenced evidence that later strengthened hazard–resource maps and the identification of priority zones for ecological restoration and climate-resilient infrastructure.

1.2.2.3.4 Tool 1: Hazard and Resource Mapping

The hazard and resource mapping tool enabled community members to visually represent their environmental and socio-economic landscape (Figure 4). Through participatory sketching, communities identified settlement clusters, farmlands, grazing areas, mangrove forests, seaweed farms, rivers, swamps, markets, schools, health facilities, water points and religious centres. Hazard features; such as recurrent flood zones, storm-surge-prone coastal segments, areas with saline intrusion, strong-wind corridors and crocodile-risk stretches along River Umba, were marked and discussed. This mapping exercise generated spatially explicit knowledge that later informed adaptation targeting and zoning.

Figure 4: Hazard and Resource Mapping in Mikoroshoni Village, Vanga Ward (Source: PCVA, 2025)

1.2.2.3.5 Tool 2: Historical Timelines

Historical timelines reconstructed multi-decadal climatic and socio-economic events including major droughts, destructive floods, storm events, pest invasions (armyworms, whiteflies and rice pests), fishery fluctuations linked to monsoon variability, severe seaweed farm losses from temperature anomalies and episodes of heightened human–wildlife conflict. Each event was discussed in terms of magnitude, social and economic impact, community responses and institutional support. Timelines revealed a pattern of intensifying climatic shocks, shortened recovery cycles and cumulative stress on natural resource systems.

1.2.2.3.6 Tool 3: Seasonal Calendars

Seasonal calendars detailed intra-annual cycles of rainfall, temperature, wind regimes, tidal patterns, agricultural operations, fishing seasons, seaweed planting and harvesting, mangrove harvesting, disease outbreaks and social events. Comparisons between historical and current seasonal behaviour revealed declining predictability of rainfall, increased occurrence of off-season rainfall, persistent dry spells and shifts in traditional fishing and seaweed farming seasons. Calendars also captured periods of heightened GBV, school absenteeism, food scarcity and increased care burdens.

1.2.2.3.7 Tool 4: Daily Clock

The daily clock exercise revealed gendered labour allocation and how climate stress reconfigures time use. Under climate-normal conditions, tasks such as farming, seaweed farming, childcare, water collection and marketing followed relatively stable patterns. Under climate stress, women and girls reported significant increases in unpaid care work and water collection time, men noted prolonged periods of income loss due to weather-inhibited fishing, and youth described increased domestic workload substitution and reduced learning hours. These insights demonstrated how climate shocks deepen time-poverty and reduce adaptive capacity.

1.2.2.3.8 Tool 5: Household Decision-Making Pile Sorting

Pile sorting exercises clarified intra-household power/authority structures under normal and climate-stress scenarios. Under normal conditions, decision-making on crop choices, livestock sales and minor household investments tended to be joint or gender-specific depending on cultural norms. Under climate stress, however, the decision space often shifted toward male dominance, particularly in decisions involving financial risk, fishing activity, asset disposal or climate-response investment, which in turn constrained women's adaptive agency and reinforced gender-based vulnerabilities.

1.2.2.3.9 Tool 6: Leaky Bucket

The Leaky Bucket tool illustrated how climate shocks disrupt household financial stability. Communities mapped income streams such as farming, fishing, seaweed sales, petty trade, remittances and casual labour, alongside expenditure flows including food purchases, health costs, water, school fees, debt servicing and transport. Climate shocks were shown to reduce inflows drastically while simultaneously raising outflows through rising food prices, repair costs and medical expenses. This widening income–expenditure gap demonstrated the structural economic vulnerability of households.

1.2.2.3.10 Tool 7: Impact Chains

Impact chain analysis depicted the cascading consequences (direct and indirect impacts) of priority hazards. For instance, prolonged drought triggered crop losses, water scarcity, livestock mortality and reduced milk intake, which in turn contributed to household food insecurity, malnutrition risk among children, increased labour migration by men, rising care burdens for women, heightened GBV risk and reduced school attendance. Similarly, strong winds disrupted fishing and seaweed farming, leading to economic losses, debt accumulation and psychological stress. This tool exposed multi-dimensional vulnerability pathways and supported the design of integrated adaptation strategies.

1.2.2.4 Step 4: Analysis of Causes of Vulnerability (Tool 8 – Vulnerability Matrix)

The analysis of underlying causes of vulnerability was undertaken through the application of the vulnerability matrix (Tool 8), which provided a structured means of examining how different climatic hazards intersect with livelihood assets and socio-ecological systems in Vanga Ward. Hazards such as drought, floods, strong winds, sea-level rise, human–wildlife conflict and agricultural pests and diseases were arranged against critical assets including rain-fed farms,

irrigated plots, mangrove forests, seaweed farms, artisanal fisheries, livestock, domestic water sources, housing structures, schools, health facilities, feeder roads and market infrastructure. Community members assigned severity ratings for each hazard–asset interaction, supported by qualitative reasoning and lived examples.

The cumulative scoring process produced a ranking that demonstrated, for example, the high susceptibility of mangrove areas and seaweed farms to combined effects of temperature stress, strong winds and changing salinity regimes; the acute vulnerability of rain-fed agriculture to prolonged drought and unpredictable rainfall; and the differentiated impacts of floods on schools and feeder roads which disrupt mobility, market access and educational continuity. This step bridged descriptive hazard analysis with deeper socio-economic and institutional determinants of vulnerability, forming the foundation for identifying priority adaptation pathways.

1.2.2.5 Step 5: Analysis of Community Action (Tool 9 – Venn Diagrams)

Analysis of existing coping and adaptation practices revealed a range of coping strategies and externally supported interventions. The Venn diagram exercise (Tool 9) enabled communities to map the institutional ecosystem influencing adaptation in the ward. Institutions such as county line departments, NGOs, community-based organisations, religious institutions, savings and credit groups, fisheries and forestry officers, COSME project structures, and private sector actors were placed on diagrams according to their perceived influence and accessibility. The spatial arrangement revealed a significant presence of institutions but varying degrees of functional support. While some organisations maintained close, supportive relationships, such as community groups collaborating with NbS and seaweed farmer associations, others operated at a distance, limiting their impact. This analysis also identified critical gaps, including insufficient extension services for seaweed and fisheries, weak linkages to financial services, limited climate information dissemination, and inadequate institutional frameworks for systematic PAD inclusion. These findings provided essential insights for designing governance and institutional strengthening elements within the Ward Adaptation Plan.

1.2.2.6 Step 6: Drawing Action from Analysis (Tool 10 – Adaptation Pathways)

The final step in the participatory diagnostic process involved translating identified vulnerabilities, exposure pathways and institutional gaps into a coherent adaptation strategy through the Adaptation Pathways tool (Tool 10). Communities revisited impact chains and vulnerability dynamics to identify feasible response strategies across short-, medium- and long-term horizons. The exercise involved structured deliberations that considered gendered labour implications, the availability of financial and natural resources, household and community capacities, and alignment with ongoing COSME and county initiatives.

Resulting adaptation pathways articulated sequenced actions for key socio-ecological systems; agriculture, water resources, mangrove and terrestrial forests, seaweed farming and fisheries, community infrastructure, early warning systems and social protection mechanisms. This included immediate resilience-building measures such as water harvesting, reinforcement of seaweed lines, community drainage clearance and household-level livelihood diversification; medium-term actions such as improved coastal protection, establishment of climate-resilient seaweed strains, expansion of irrigation infrastructure and strengthening local governance structures; and long-term interventions such as large-scale mangrove rehabilitation, institutionalization of community-based climate monitoring systems, and improvement of rural infrastructure. These pathways formed the backbone of the Vanga Ward Adaptation Plan, linking diagnostic insights to strategic action.

1.2.3 Household Survey

A structured household survey was administered to 295 households in the study area to provide quantitative evidence complementing qualitative findings. Sampling was designed to include households across diverse livelihood portfolios and different vulnerability categories, including female-headed households, large households with high dependency ratios, households with PAD members, and those primarily reliant on climate-sensitive livelihoods such as rain-fed agriculture, artisanal fishing and seaweed farming. Primary respondents were predominantly women (approximately 65 %), ensuring robust capture of intra-household dynamics, care burdens, access to services and lived experiences of climate stress.

The survey collected data on demographic structure, education, disability status, land tenure and resource access, livelihood contributions, income diversification, exposure to recent climatic shocks, losses and damages, coping strategies, food security indicators, institutional linkages and participation in social protection and savings groups. Data were collected digitally using KoboToolbox to ensure consistency and reduce transcription errors. These data provided a statistically grounded basis for validating PCVA findings and understanding patterns of vulnerability at household level across gender, age and livelihood categories.

1.2.4 Current Mangrove and Forest Health Assessment

To assess the condition of key coastal and terrestrial ecosystems supporting nature-based adaptation in Vanga, a remote sensing analysis was conducted using Sentinel-2 imagery at 10-metre spatial resolution. Imagery underwent standard pre-processing steps including radiometric correction and cloud masking before the Normalized Difference Vegetation Index (NDVI) was computed using the formula.

$$\text{NDVI} = \frac{NIR - RED}{NIR + RED} \qquad \text{Equation 1}$$

Where RED is visible red reflectance, and NIR is near infrared reflectance.

NDVI values were categorised into low, moderate and high vegetation vigor classes. Low NDVI signatures corresponded to degraded or stressed vegetation, often reflecting factors such as saltwater intrusion, wood extraction pressure, sedimentation and hydrological alteration. High NDVI zones served as ecological benchmarks, indicating healthier vegetation stands. These spatial patterns were subsequently compared with PCVA hazard–resource maps and field observations to identify priority sites for ecological restoration, mangrove replanting, mangrove hydrological rehabilitation and sustainable forest management within the ward.

1.2.5 Current and Future Climate Projections

Climate projections relevant to Vanga were developed using downscaled county-level analyses supplemented by remote sensing and historical observational datasets. Daily rainfall Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS), Actual Evapotranspiration (AET), Potential Evapotranspiration (PET), (from ERA5-Ag) and temperature variables (minimum, mean, maximum and Land Surface Temperature (LST) from Landsat and ERA5-Ag) were extracted through Google Earth Engine after spatial boundaries had been validated in QGIS. The resulting datasets were aggregated into annual totals or means and subsequently analysed using Python (Colab) employing pandas, numpy, scikit-learn and matplotlib.

Linear regression models were fitted to each climatic variable to estimate historical trends and project future changes to 2035 and 2045. Temperature projections incorporated Intergovernmental Panel on Climate Change (IPCC) AR6-aligned warming increments for East Africa's coastal belt, applying approximately +0.02°C per year for the moderate-emission

scenario and +0.035°C per year for high-emission trajectories. Rainfall projections followed empirically derived trends, showing increasing variability, changes in seasonal distribution and heightened likelihood of intense episodic rainfall events. These climatic insights were interpreted in relation to Vanga's hazard profile, reinforcing the need for enhanced water security, climate-resilient agricultural practices, improved coastal and marine ecosystem management, stronger infrastructure planning and expanded early warning systems.

1.2.6 Data Analysis and Reporting

Quantitative data from the household surveys were cleaned, coded and analyzed using Microsoft Excel and Python. Analysis focused on frequencies, proportions, cross-tabulations and simple correlations across gender, age group, livelihood type and vulnerability status. These results provided empirical confirmation of patterns emerging from qualitative insights.

Qualitative data from FGDs, PCVA exercises and transect walks were transcribed and thematically coded to organise evidence under key domains such as hazard exposure, livelihood impacts, gender dynamics, coping strategies, institutional barriers and opportunities. Outputs from PCVA tools; maps, timelines, calendars, clocks, matrices and impact chains, were digitized and integrated with NDVI layers and climate trend visualizations to produce synthesized analytical outputs. Triangulation across all sources ensured internal consistency and improved the robustness of the final interpretation. Draft findings were compiled for review by Plan International Kenya in line with expected deliverables.

1.2.7 Validation

The draft findings were presented in county-level validation workshops held in Kilifi and Kwale on 20th and 21st November 2025 respectively. These sessions provided a platform for county technical departments, CCPC/WCCPC members, civil society, COSME partners and community representatives to review the PCVA results and refine the draft adaptation strategies. At ward level, stakeholders from Vanga reviewed hazard rankings, adaptation priorities and institutional roles, and provided further refinement on feasibility, gender inclusion, sequencing and resource implications. Inputs from these processes were incorporated into the final Ward Adaptation Plan to ensure scientific accuracy, community legitimacy and institutional alignment.

1.3 Linkages to other planning processes

At both the National, County and Ward Levels, several other planning processes exist to which the WAP is linked. These include the National Adaptation Plan 2015-2030, the Kwale County Adaptation Plan 2023-2032 and Vanga Ward level Participatory Vulnerability Climate Risk Assessment (PVCA) conducted in 2023. The NAP is Kenya's first plan on adaptation and demonstrates the commitment to operationalize the National Climate Change Action Plan by mainstreaming adaptation across all sectors in the national planning, budgeting and implementation processes. The mainstreaming approach recognizes that climate change is a cross-cutting sustainable development issues with economic, social and environmental impacts. Similarly, the CAP aligns with the NAP in terms of content and reporting focusing on Kwale County circumstances, climate hazards and adaptation actions which are largely derived from PCVAs for the 20 wards. The WAP adaptation plan is prepared considering both the NAP and CAP and aligning adaptation actions accordingly as informed by the village focused PCVAs.

2 Chapter 2: Coordination of Adaptation Actions

2.1 Introduction

Effective coordination converts adaptation intent into measurable resilience outcomes. In Vanga Ward, climate risks cut across water security, agriculture, coastal ecosystems, health, and infrastructure, and they are experienced unevenly by households, with women and other vulnerable groups often bearing disproportionate burdens. Translating the Ward Adaptation Plan into practice therefore requires a governance architecture that aligns local priorities with county programming and national direction, while ensuring timely access to finance, technical support, and decision-useful climate information. This chapter sets out that architecture and links it explicitly to Kenya's National Adaptation Plan (NAP) 2015–2030. The coordination approach is designed to be both horizontal, fostering collaboration among village institutions, civil society, private actors, and technical officers inside the ward and vertical embedding ward priorities in county plans and budgets and aligning them with NAP strategic actions in water resources management, climate-smart agriculture, coastal zone management, disaster risk reduction and early warning, health systems resilience, and ecosystem restoration. The result is a coherent set of institutions and processes through which roles are clarified, duplication and gaps are minimized, and gender-responsive, locally led adaptation is resourced, implemented, and monitored over time.

2.2 Current Institutional Arrangements

Coordination presently operates across three interlinked tiers. At national level, policy direction and technical backstopping are provided through the Climate Change Directorate and the National Climate Change Council under the Climate Change Act (2023), with operational guidance through the NAP and the National Climate Change Action Plan (NCCAP) 2023–2027. Specialized agencies complement this architecture: the Kenya Meteorological Department issues forecasts and impact-based warnings essential for agricultural planning, coastal safety, and disaster preparedness; the National Drought Management Authority convenes drought early warning and response mechanisms, including triggers for social protection; the Kenya Forest Service and Kenya Wildlife Service support ecosystem restoration and human–wildlife conflict management relevant to the ward's terrestrial and coastal interfaces; and the National Environment Management Authority administers environmental safeguards for adaptation investments. These agencies typically interact with Vanga via county structures, with ad hoc escalation during seasonal extremes.

At county level, Kwale County Government acts as the bridge between national strategy and local action. The County Climate Change Act (2022) establishes the County Climate Change Unit to coordinate climate programming and operationalize the County Climate Change Fund (CCCF). Climate mainstreaming is progressively embedded in the County Integrated Development Plan (CIDP) 2023–2027 and sectoral plans, enabling the Departments of Water; Agriculture, Livestock and Fisheries; Roads and Public Works; Health; and Environment and Forestry to integrate resilience criteria into annual work plans. County coordination platforms such as the County Steering Group and the Disaster Management Committee manage multi-hazard preparedness and response and are increasingly used to plan anticipatory actions informed by seasonal outlooks. Partnerships with non-governmental organizations and development partners are convened through county-led forums so that external programming in Vanga is harmonized with public investments and avoids duplication.

At ward and community level, implementation energy is concentrated. The Ward Administrator ensures administrative linkage to county directorates, while the Ward Climate Change Planning Committee (WCCPC) provides an inclusive forum for agenda setting, investment prioritization, and progress review. The WCCPC comprises local administration, technical extension officers, Beach Management Units, Water User Associations, Community-

Based Organizations, women's and youth representatives, and respected elders. This composition mirrors the NAP's emphasis on locally led adaptation and social inclusion. Issue-specific structures such as water user committees, disaster volunteer teams, and environmental stewardship groups execute day-to-day actions: maintaining boreholes and storage, coordinating flood preparedness, overseeing mangrove rehabilitation, and managing communal assets. Historically, coordination among these groups has relied on personal networks; the WCCPC now provides a structured convening mechanism for information sharing, joint planning, and accountability across villages and sectors.

2.3 Relevant Policies

The policy environment guiding coordination is robust and multi-scalar. The Climate Change Act (2023) mandates climate integration across public planning, establishes coordination bodies, and provides the legal basis for devolved climate finance. The NAP 2015–2030 defines national priorities for adaptation and sets out sectoral action areas that are directly germane to Vanga: strengthening integrated water resources management and water supply resilience; scaling climate-smart agriculture and resilient livestock systems; safeguarding health through surveillance and risk communication; climate-proofing infrastructure and human settlements; protecting and restoring forests, mangroves, and coastal buffers; and institutionalizing risk-informed development through early warning and disaster risk reduction. The NCCAP 2023–2027 operationalizes these priorities into programs and financing pathways, notably the Financing Locally Led Climate Action (FLLoCA) program that channels performance-based climate grants to county and ward levels.

Sub-nationally, the Kwale County Climate Change Act (2022) and the County Climate Change Action Plan (2023–2027) provide an enabling framework for ward proposals to access the County Climate Change Fund and align with county targets on water security, resilient agriculture, coastal and riparian restoration, and climate-proofed infrastructure. Sector-specific policies reinforce these mandates: the Water Act and National Water Policy embed community-based water governance and integrated water resources management; the Agricultural Sector Transformation and Growth Strategy promotes diversified, climate-smart production and value addition; the National Disaster Risk Management policy architecture advances anticipatory action and local preparedness; the Wildlife Conservation and Management Act provides mechanisms for human–wildlife conflict mitigation and compensation; and national gender and social inclusion policies require equitable participation and benefit sharing. Together, these instruments establish the principles, standards, financing mechanisms, and safeguards that the Vanga coordination system adopts, thereby ensuring consistency with the NAP while tailoring execution to place-based realities.

2.4 Proposed Coordination Arrangements

The proposed arrangement consolidates existing strengths and introduces defined interfaces and routines to meet NAP expectations for coherence, subsidiarity, and learning. At the ward–county interface, a designated County Climate Change Officer will formally serve as liaison to the WCCPC, attending scheduled meetings, sharing programmatic updates, and translating ward priorities into departmental plans and CCCF pipelines. Quarterly progress notes prepared by the Ward Administrator and endorsed by the WCCPC will flow to the County Climate Change Unit, documenting outputs, challenges, gender-disaggregated beneficiary data, safeguards compliance, and next-quarter needs. This reporting cadence embeds a performance culture compatible with FLLoCA and county M&E systems.

Within the ward, the WCCPC will convene a biannual Vanga Climate Stakeholder Forum that aggregates all active and potential actors; county departments, national agencies operating locally, NGOs, CBOs, BMUs, WUAs, private firms, research institutions, religious organizations,

and microfinance and insurance providers. The forum will map who is doing what, where, and with which resources; agree on technical standards; broker complementary roles across villages; and surface implementation bottlenecks for resolution. Between forums, the WCCPC will chair monthly coordination sessions focusing on near-term execution tasks, including the sequencing of procurements, community mobilization, safeguards checks, and integration of seasonal forecasts into operational plans.

Project implementation will be delegated to time-bound, task-specific teams constituted as sub-committees of the WCCPC, each blending community oversight with technical supervision. A water supply team, for example, will include the relevant Water User Committee, a county water engineer, village leadership, and, where applicable, an NGO or private contractor representative. An ecosystem restoration team will draw on Kenya Forest Service and BMU expertise, women's groups managing nurseries, and local conservation champions. These teams will meet frequently during active phases, maintain work logs and financial records, and furnish concise monthly updates to the WCCPC.

Because Vanga faces recurrent hazards, adaptation coordination will be explicitly integrated with disaster risk management. The WCCPC will either serve as, or substantially overlap with, the Ward Disaster Management Committee to ensure seamless transition between long-term resilience building and acute event response. Protocols will define how KMD advisories are received, interpreted, and disseminated through village communication trees; how NDMA triggers activate anticipatory measures and social protection linkages; and how ward teams coordinate with county emergency units for water trucking, evacuation, or health campaigns. This risk-informed approach is consistent with NAP priorities on early warning systems and the mainstreaming of DRR into development planning.

Flexibility and learning are designed into the system. An annual reflection, co-convened by the County Climate Change Unit and the WCCPC, will review results against NAP-aligned indicators, document lessons, and adjust institutional arrangements or operating procedures. As national guidance, data systems, or financing windows evolve, the ward will adapt its coordination protocols, ensuring continuity with national direction while preserving local ownership.

2.5 Implementation Roles

Effective implementation of the Ward Adaptation Plan requires clear role delineation among all stakeholders to ensure that adaptation measures are executed efficiently, inclusively, and in line with Kenya's National Adaptation Plan (2015–2030). Table 1 below outlines the mandates, decision-making authority, and accountability mechanisms of the key institutions and actors engaged in rolling out adaptation interventions in Vanga Ward. The WAP explicitly mainstreams Gender Equality and Inclusion (GEI) across all stages of planning and implementation and establishes strong coordination linkages with health, gender, and protection actors to guarantee continuity of essential Sexual and Reproductive Health and Rights (SRHR) and GBV prevention and response services during climate shocks.

Table 1: Institutional Roles in Implementation of Adaptation Actions

Institution/Actor	Key Roles and Responsibilities
Ward Climate Change Planning Committee (WCCPC)	Lead planning, coordination, and prioritization of ward-level adaptation actions; mobilize community participation; ensure gender and social inclusion in project design and implementation.
Kwale County Government Departments	Provide technical oversight, finance, and sectoral expertise (Water, Agriculture, Livestock, Fisheries,

Institution/Actor	Key Roles and Responsibilities
	Roads, Health, Environment). Integrate ward priorities into departmental annual workplans and budgets.
National Drought Management Authority (NDMA)	Lead drought monitoring, early warning, and response; integrate Vanga into national drought early warning systems and coordinate emergency relief when thresholds are reached.
Kenya Meteorological Department (KMD)	Provide localized weather and climate forecasts, early warnings, and advisories for decision-making in agriculture, fisheries, and disaster preparedness.
Kenya Forest Service (KFS)	Offer technical guidance on reforestation, afforestation, and mangrove restoration; support tree nursery establishment and monitoring.
Kenya Wildlife Service (KWS)	Manage and mitigate human–wildlife conflicts; support establishment of water points for wildlife and training of community wildlife scouts.
National Environment Management Authority (NEMA)	Review and approve Environmental Impact Assessments (EIAs) for adaptation projects; ensure adherence to environmental safeguards.
Development Partners (UNDP, GCF, World Bank, EU, FAO, IFAD)	Provide technical assistance, capacity building, and climate finance to support implementation of local adaptation priorities; promote mainstreaming of climate resilience in development investments and infrastructure.
Non-Governmental Organizations (NGOs) and CSOs (Plan International, WWF-Kenya, IUCN, Care Kenya, World Vision Kenya, Base Titanium Limited)	Support implementation of community-based adaptation projects; promote gender-responsive climate resilience initiatives; facilitate training and awareness creation on nature-based solutions (NbS) and livelihood diversification.
Community-Based Organizations (CBOs) and Self-Help Groups	Implement localized adaptation initiatives such as mangrove restoration, water harvesting, and livelihood diversification; mobilize local labor and indigenous knowledge; ensure equitable participation in project benefits.
Private Sector and Cooperatives	Promote investment in climate-smart agriculture, renewable energy, and insurance products; integrate environmental sustainability in business operations; collaborate in corporate social responsibility (CSR) initiatives that support community resilience.
Academic and Research Institutions (KMFRI, KEMFRI, Universities)	Generate and disseminate climate and ecological data; support participatory research and innovation for adaptation technologies; build capacity of local institutions on ecosystem monitoring and early warning systems.

3 Chapter 3: Situation analysis of climate vulnerability

3.1 Introduction

The ward is increasingly experiencing multiple, interlinked, and climate-induced hazards whose intensity, duration, and frequency have grown over time. Findings from the Participatory Climate Vulnerability and Capacity Assessment (PCVA, 2025) indicate that the community is recurrently affected by drought, floods, sea level rise, pests and diseases, strong winds, environmental degradation, and emerging human–wildlife conflicts (HWCs). These hazards have progressively disrupted ecosystem balance, altered natural resource functions, and reshaped traditional livelihood systems across Vanga, Jimbo, Kiwegu, Jasini, Perani, Sega, and Mwalewa areas. The assessment employed a combination of community-based knowledge, historical trend analysis, and scientific projections to establish the frequency, intensity, and spatial extent of each hazard. Vulnerability was analyzed across the four sampled villages i.e., Mikoroshoni, Sega D, Ngweneni, and Mwalewa C, based on two key parameters: exposure to climatic stressors and adaptive capacity at household and institutional levels.

The results reveal that women, widows, elderly persons, persons abled differently (PAD), and low-income households bear the greatest burden of climate-related impacts. Their heightened vulnerability stems from dependence on climate-sensitive natural resources, limited access to financial and productive assets, and increased unpaid labour associated with caregiving and resource collection during crisis periods. Spatial analysis further shows that sea level rise is most severe in coastal zones of Vanga, Jimbo, Kiwegu, and Jasini; drought affects mangrove ecosystems, farmlands, swamps, and boreholes throughout the ward; while flooding is concentrated in low-lying terrestrial areas of Perani, Sega, Vanga, and Kiwegu, as well as along the River Umba and coastal settlements such as Botola and Mwalewa. Understanding these spatial and demographic dynamics provides a crucial evidence base for prioritizing adaptation interventions aimed at minimizing exposure, reducing vulnerability, and enhancing the long-term resilience of communities and ecosystems in Vanga Ward.

3.2 Current climate trends

Vanga Ward experiences a tropical coastal climate characterized by bimodal rainfall and warm temperatures year-round. The current climatic conditions in Vanga Ward, based on the baseline analysis under the SSP2–4.5 scenario, indicate a moderately wet environment with annual rainfall averaging 1,050.1 mm, against a potential evapotranspiration (PET) of 1,478.6 mm and an actual evapotranspiration (AET) of 1,675.9 mm (Figure 6). The consistently higher PET values compared to rainfall demonstrate that atmospheric water demand exceeds precipitation input, predisposing the area to periodic moisture deficits and hydrological stress, especially during the dry months. This condition is typical of coastal agro-ecological systems where elevated temperatures and seasonal rainfall variability interact to create short-lived wet periods followed by extended dry spells.

Temperature analysis under the same scenario (Figure 7) shows a mean air temperature of 25.8°C and a land surface temperature (LST) of 34.4°C, indicating the presence of considerable surface heating across terrestrial landscapes. The thermal intensity is influenced by low vegetative cover, exposed soils, and increasing land-use conversion, particularly settlement expansion. The thermal and hydrological profiles collectively indicate that the current climate system in Vanga Ward is characterized by moderate rainfall, elevated evapotranspiration, and persistent soil moisture stress, with implications for agricultural productivity, vegetation health, and freshwater availability.

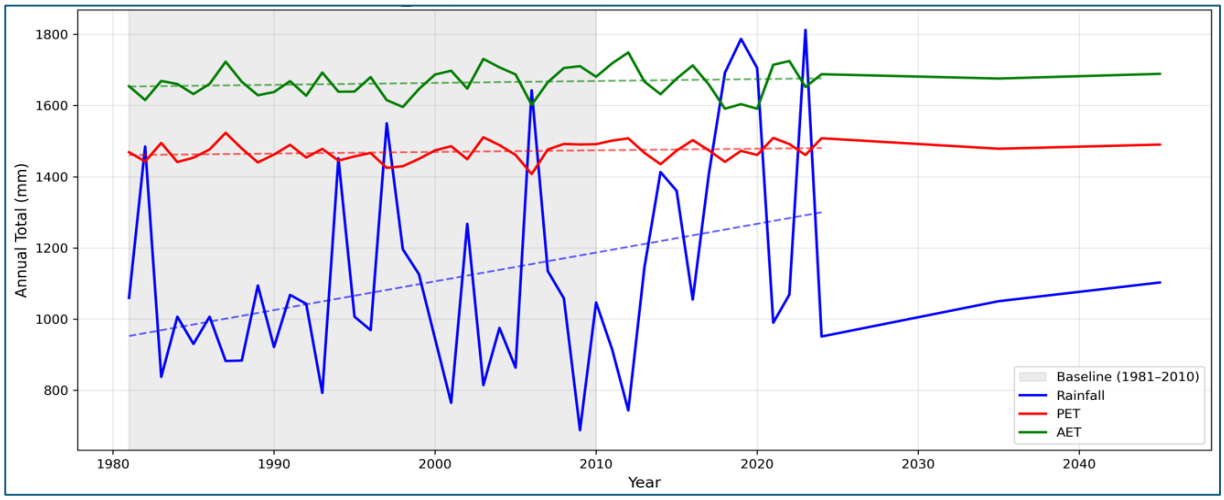

Figure 5: SSP2 4.5 Rainfall, AET and PET Projections (Source: Chirps database and ERA5-Ag, 2022)

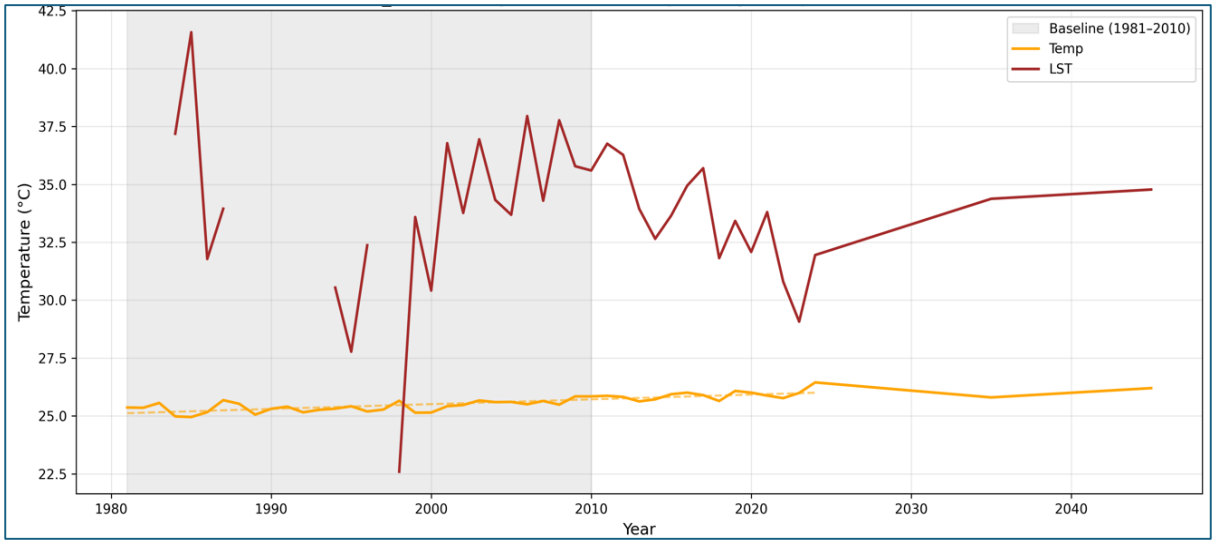

Figure 6: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.3 Future climate trends

Projected climate conditions under the high-emission scenario (SSP5–8.5) indicate that Vanga Ward will experience slightly wetter but considerably warmer conditions relative to the baseline period. Rainfall is projected to increase from 1,102.6 mm to 1,155.1 mm, PET from 1,490.3 mm to 1,510.8 mm, and AET from 1,689.2 mm to 1,712.4 mm (Figure 8). Although total precipitation is expected to rise, PET continues to outpace rainfall, signifying persistent evaporative demand and a potential decline in effective water availability. This trend implies that even under wetter conditions, hydrological stress and soil moisture deficits will likely persist, particularly during dry seasons.

Future temperature projections further emphasize the warming trajectory. The mean air temperature is expected to rise from 25.9°C to 26.3°C, while the LST increases significantly from 36.9°C to 37.3°C (Figure 9). The strong surface heating effect reflects enhanced radiative forcing associated with greenhouse gas accumulation and local land degradation processes. Such

temperature increases are expected to intensify evapotranspiration rates, accelerate crop water loss, and increase human and livestock heat stress.

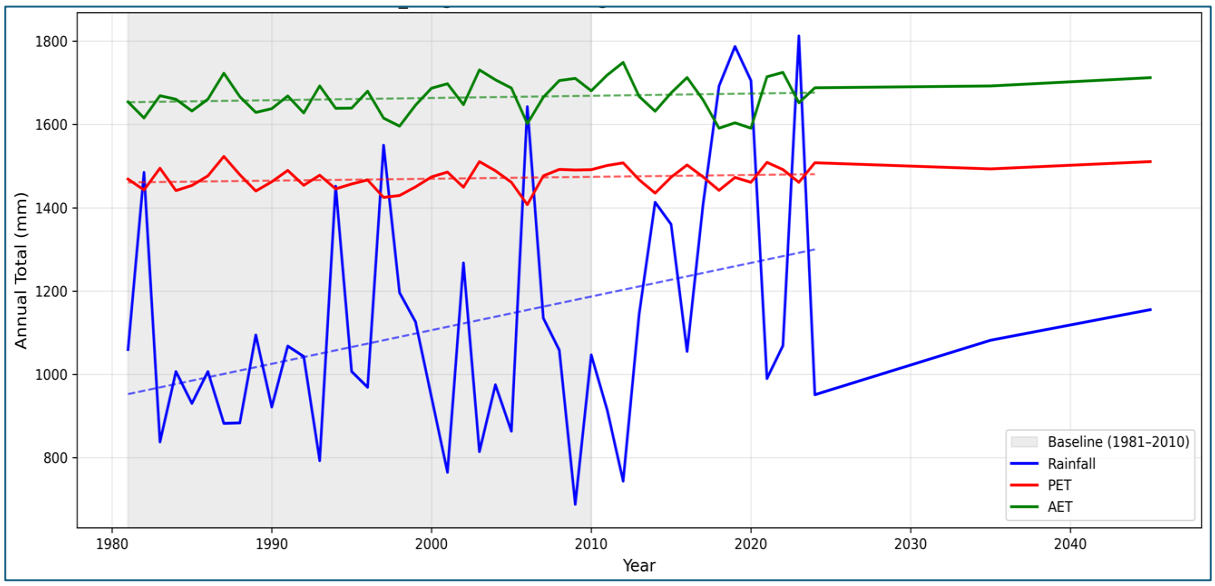

Figure 7: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022)

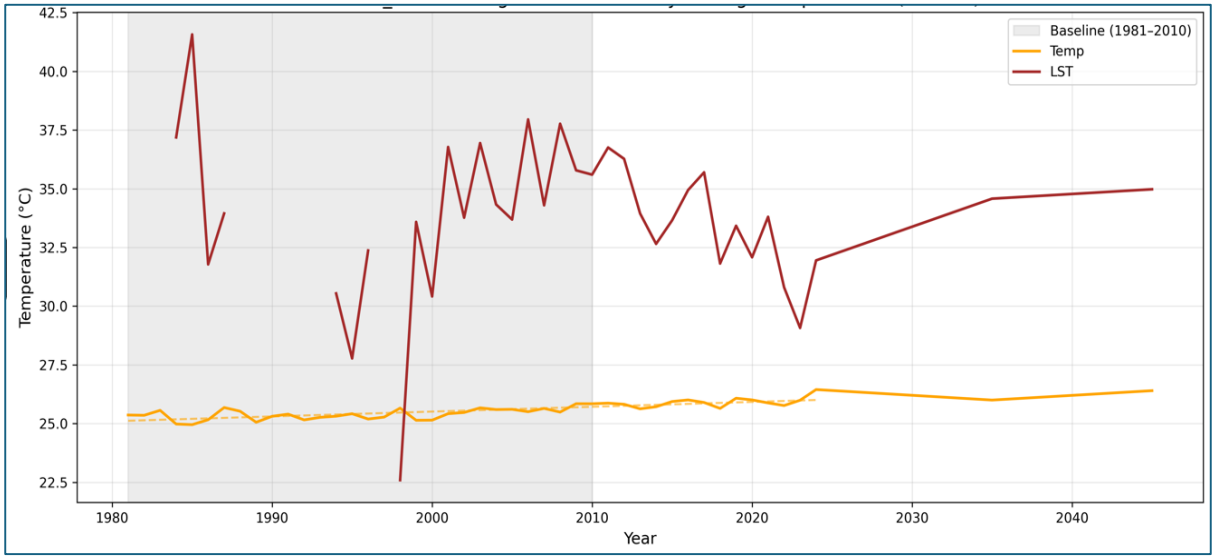

Figure 8: SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.4 Climate hazards and vulnerability

3.4.1 Drought

Drought remains the most persistent and economically devastating climate hazard in Vanga Ward, placing sustained pressure on both terrestrial and coastal ecosystems. Historical climate records from 1971 to 2025 indicate that major drought events occur approximately once every four years, with notable severe episodes recorded in 1971, 1992, 2005–2007, 2017, and 2024 (Figure 10). Over time, the frequency, duration, and intensity of these dry spells have increased, reflecting a marked decline in rainfall reliability and prolonged inter-seasonal dry periods. This trend is closely linked to delayed rainfall onset, early cessation of the long rains, and rising evapotranspiration rates, all of which have contributed to shorter growing seasons and recurrent crop failures.

Spatially, vulnerability is not uniform across the ward. Mwalewa C (vulnerability score 73) and Mikoroshoni (72) exhibit the highest exposure levels due to their heavy dependence on rain-fed agriculture, proximity to degraded forest ecosystems, and limited alternative water sources. Ngweneni (63) follows closely, while Sega D (35) experiences comparatively moderate impacts owing to better access to shallow wells and diversified livelihoods (Figure 11). Households situated on higher, less vegetated grounds experience more severe water scarcity, whereas communities near low-lying mangrove and estuarine zones are indirectly affected through declining water tables and saline intrusion into wells.

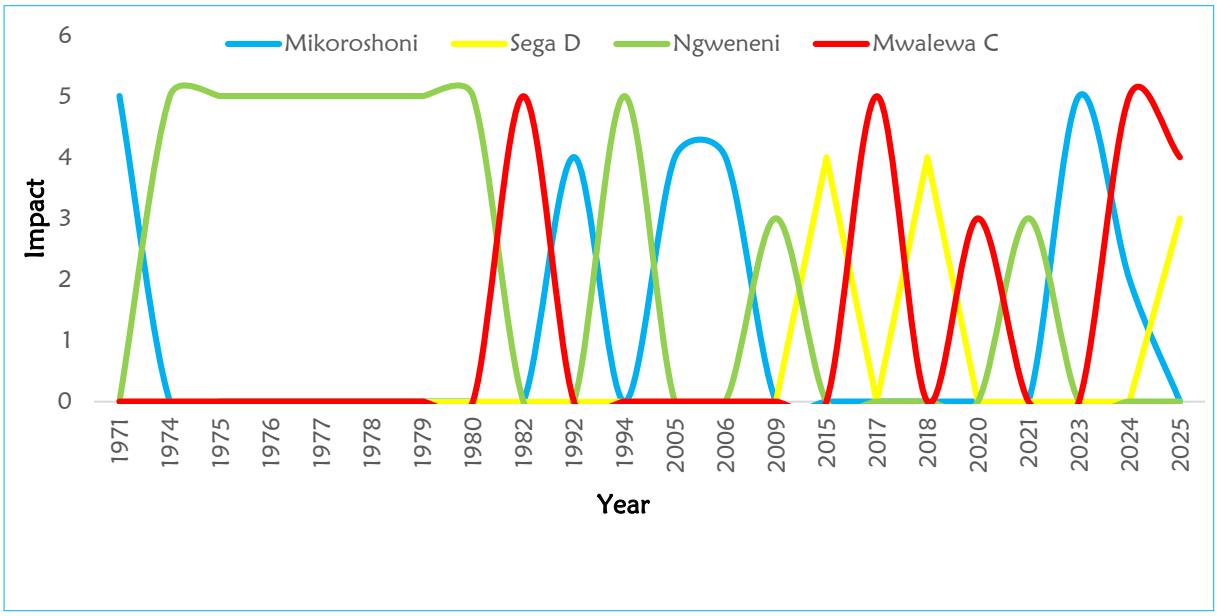

Figure 9: Drought prevalence historical trend (Source: PCVA, 2025)

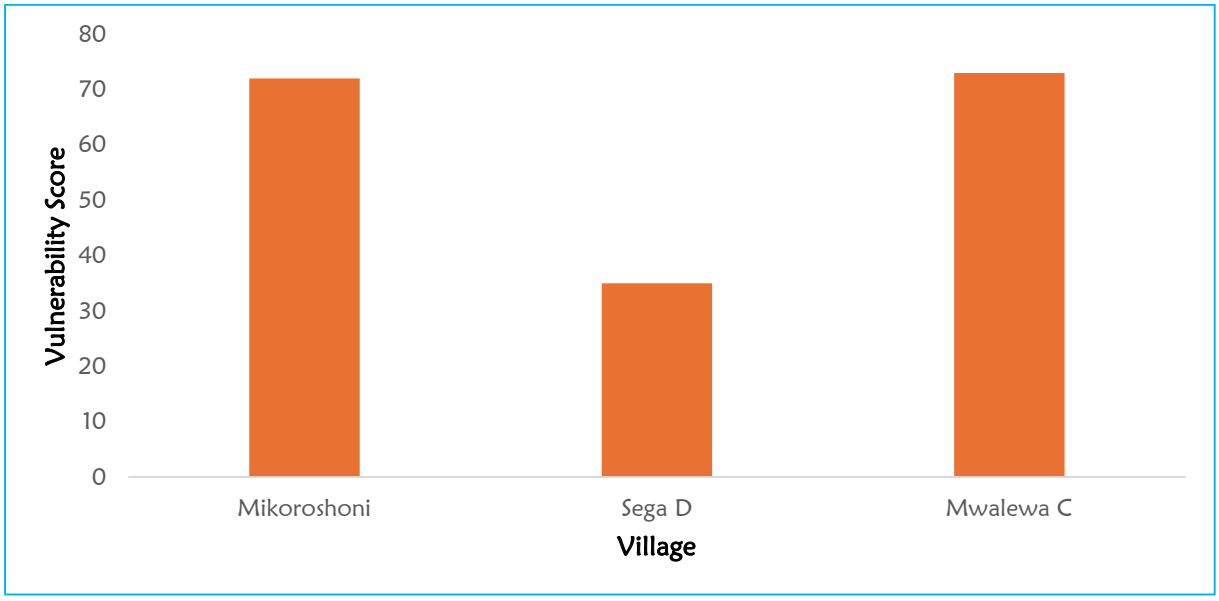

Figure 10: Drought vulnerability score in Vanga Ward (Source: PCVA,2025)

Natural resources form critical livelihood assets that exhibit varying degrees of vulnerability. The Gonja and Kaya Sega forests, for instance, have experienced notable canopy loss and drying of undergrowth species due to prolonged water stress, making them less effective as local microclimate regulators and biodiversity refuges. Gonja Forest shows notable seasonal variation

in vegetation vigor and canopy density. During the dry season, NDVI values range from 0.15 to 0.48, indicating moderate stress likely caused by limited rainfall, higher temperatures, and reduced soil moisture. In the wet season, NDVI values increase to 0.24–0.50, reflecting improved vegetation health and regeneration due to enhanced water availability and nutrient influx (Figure 12).

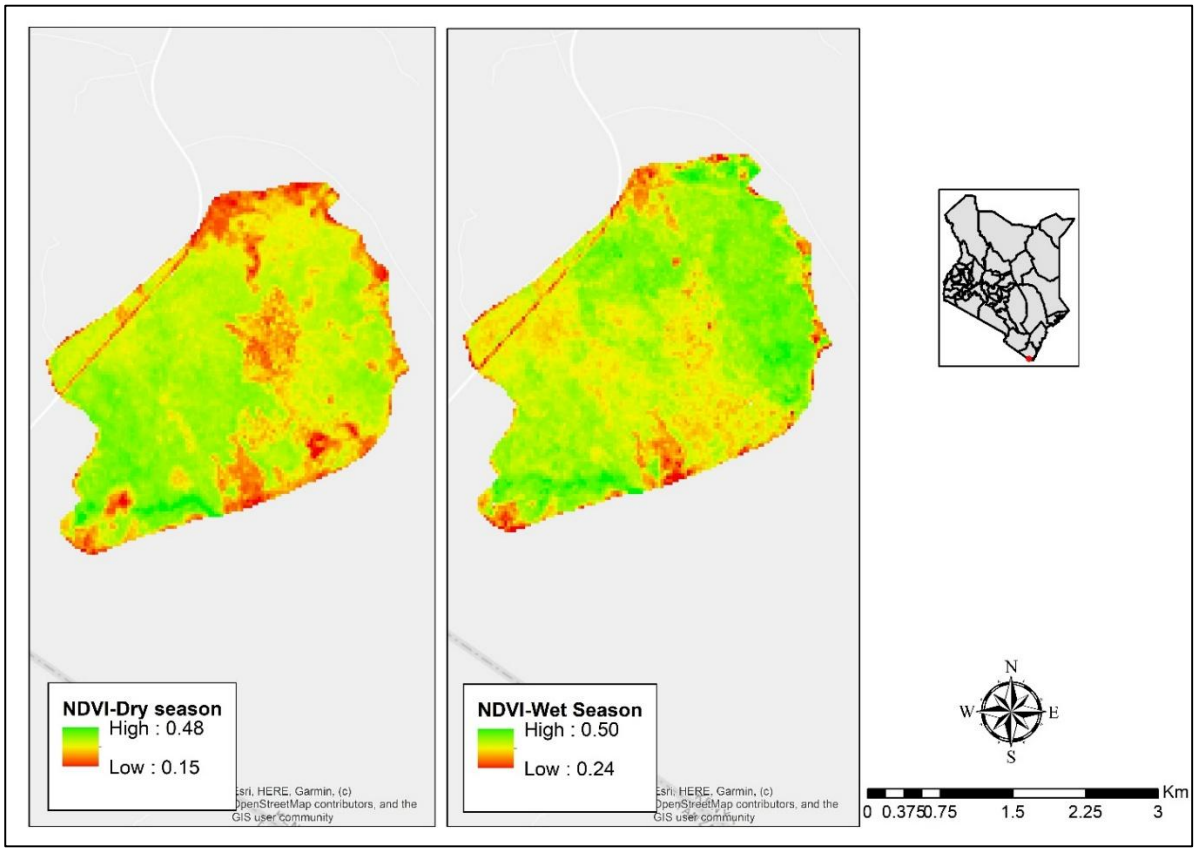

Figure 11: Gonja Forest NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2025)

Mangrove forests, particularly along the Mkurumudzi estuary, show stunted regeneration and increased seedling mortality linked to soil desiccation and hypersalinity. Similarly, seaweed farms, which depend on stable water temperatures and nutrient exchange, are periodically abandoned during extreme droughts when sea surface temperatures rise and salinity levels exceed optimal growth thresholds. During the dry season, NDVI values range from 0.12 to 0.48, indicating moderate vegetation stress caused by high salinity, reduced freshwater inflow, and elevated evapotranspiration (Figure 13). In the wet season, NDVI values generally rise to 0.14–0.55, reflecting improved vegetation health and regeneration due to increased rainfall and nutrient availability. Nevertheless, lower NDVI values observed from Majoreni to Vanga during the wet season suggest localized vegetation stress caused by sedimentation and flooding, which can smother mangrove roots, reduce oxygen availability, and hinder regeneration. These patterns underscore the sensitivity of mangrove ecosystems to seasonal climatic fluctuations, where heightened rainfall intensity and sediment loads associated with climate change may accelerate degradation and reduce the resilience of mangroves along this coastal stretch.

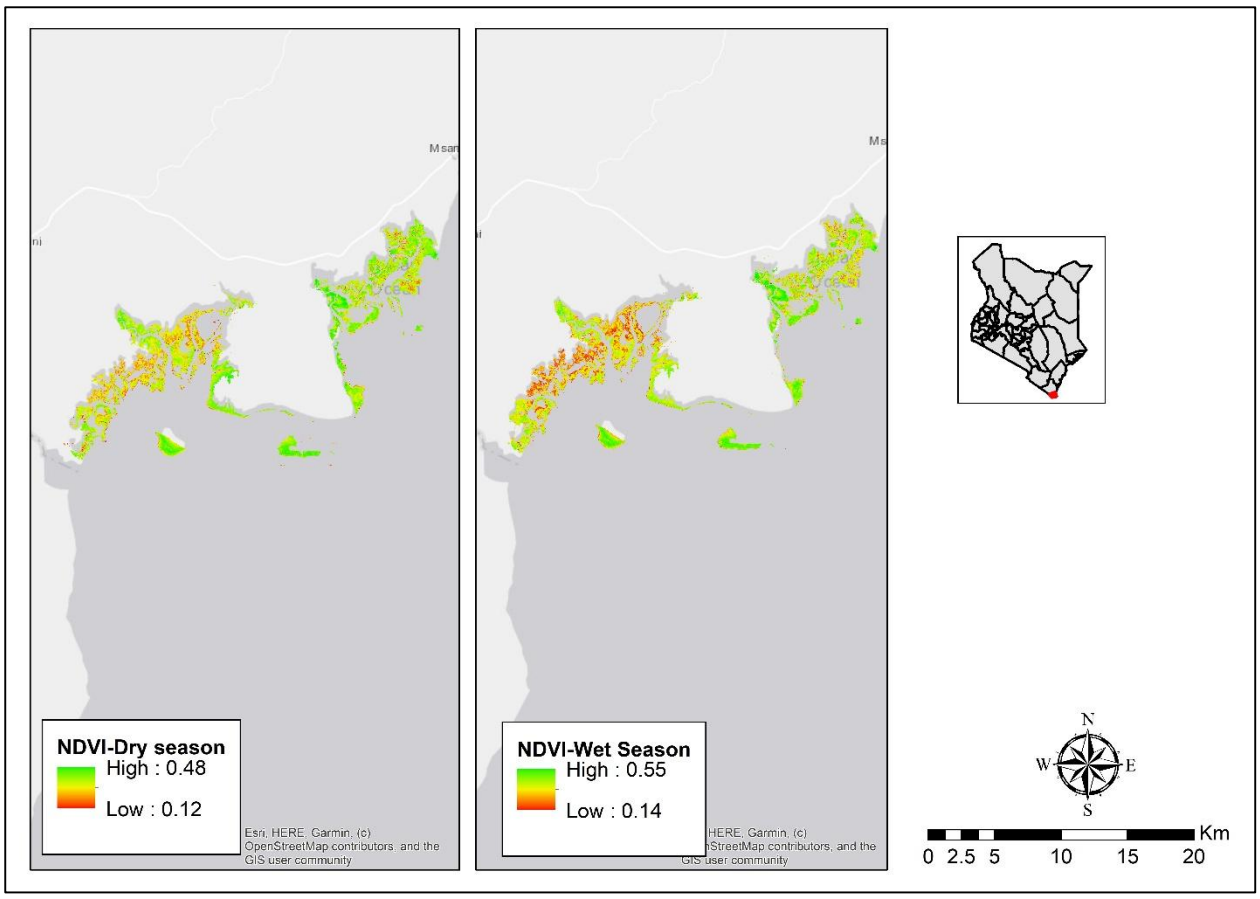

Figure 12: Mangrove Forest NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2025)

At the community level, drought impacts are strongly gendered and socially differentiated. Women and youth are disproportionately burdened as they bear the primary responsibility for water collection, fuelwood gathering, and household food provisioning. During prolonged drought periods, women in Mwalewa C and Mikoroshoni travel up to 5–8 kilometres daily in search of potable water, reducing their time for income-generating activities and increasing exposure to heat-related health risks. Men, on the other hand, experience livelihood disruptions due to reduced agricultural productivity and loss of income from fishing and charcoal trade. Elderly persons and persons abled differently (PAD) face heightened vulnerability as they have limited mobility and adaptive capacity.

3.4.1.1 Exposure to drought

Exposure to drought in Vanga Ward is extensive and cross-cutting, affecting both ecological and livelihood systems across all four villages. The most drought-prone hotspots include mangrove ecosystems in Vanga, Jimbo, Kiwegu, and Jasini, where prolonged dry spells lower tidal water exchange and increase soil salinity, leading to poor mangrove regeneration and reduced nursery function for fish species. Farmlands across the entire ward are chronically exposed, as erratic rainfall and declining soil moisture severely reduce crop yields, particularly for maize, cassava, and horticultural crops. The River Umba, which serves as a key freshwater source for both irrigation and domestic use, experiences diminished flow during extended droughts, intensifying competition for water and triggering conflicts among users. Similarly, Gonja Forest and Kaya Sega Forest are highly exposed due to drying of understorey vegetation and reduced canopy cover, which accelerates soil erosion and loss of forest biodiversity. Swamp ecosystems such as Phuni, Magangani, and Kagera also exhibit high exposure, with shrinking wetland areas leading to loss of grazing and water retention capacity. Moreover, boreholes throughout the ward face

depletion and salinization as groundwater tables recede, forcing communities to depend on unsafe or distant water sources. Livestock exposure is particularly severe in Sega Unit and the Vanga section, where prolonged scarcity of pasture and water leads to increased mortality rates, reduced milk production, and recurrent livelihood losses. This wide-ranging exposure underscores the interconnectedness of drought impacts on both natural and human systems in Vanga Ward.

3.4.1.2 Socio-Demographic Vulnerability to Drought

Vulnerability to drought in Vanga Ward exhibits marked socio-demographic disparities shaped by gender, age, and livelihood dependence. Women, particularly those heading households, are disproportionately impacted as they shoulder the primary burden of securing water, food, and fuelwood; during prolonged dry spells, they walk longer distances to reach depleted boreholes or the River Umba, heightening their physical and economic strain. Youth, especially those engaged in fishing, casual labour, and seaweed farming, experience significant livelihood disruptions as water scarcity and heat stress reduce productivity and income opportunities. Elderly persons and persons abled differently (PAD) remain highly vulnerable due to limited mobility and reliance on others for access to water and food resources. In adaptive terms, only a small fraction of households, mainly male-headed or those with diversified incomes from trade, livestock, or formal employment, are coping effectively with recurrent droughts, while the majority of smallholder farmers and seaweed cultivators are declining, facing shrinking yields and deteriorating living conditions. A few households supported by ongoing interventions, such as water harvesting and mangrove restoration programs, demonstrate improving resilience, adopting practices that stabilize livelihoods despite the increasing drought frequency. This uneven capacity to withstand and recover from drought underscores the need for gender-responsive and socially inclusive adaptation strategies within the ward.

3.4.2 Floods

Flooding remains one of the most recurrent and disruptive climate hazards in Vanga Ward, predominantly associated with intense rainfall episodes driven by the Intertropical Convergence Zone (ITCZ) and periodic El Niño. Historical flood records between 1960 and 2025 indicate major flood occurrences in 1960, 1997, 2017, 2022, and 2024, translating to an average recurrence interval of approximately one significant flood every 13 years (Figure 14). However, the frequency and magnitude of floods have intensified in recent decades due to increased rainfall variability, catchment degradation, and expansion of settlements into flood-prone zones. Vulnerability analysis shows that Mwalewa C (score 60) experiences the highest exposure, followed by Ngweneni (46) and Sega D (41), reflecting both biophysical sensitivity and socio-economic fragility (Figure 15).

The effects of floods on natural and livelihood resources are widespread and multifaceted. Gonja Forest and Kaya Sega Forest experience soil erosion, surface runoff accumulation, and uprooting of understorey vegetation, weakening their ecological function in stabilizing slopes and regulating water flow. Mangrove forests along the coastal fringe, especially in Vanga, Kiwegu, and Jimbo, face prolonged inundation and sediment deposition, which smother pneumatophores and hinder natural regeneration, ultimately reducing their capacity to protect shorelines and support fisheries. Likewise, seaweed farms situated in shallow intertidal zones are severely affected as turbulent floodwaters dislodge seaweed lines and alter salinity levels essential for optimal growth.

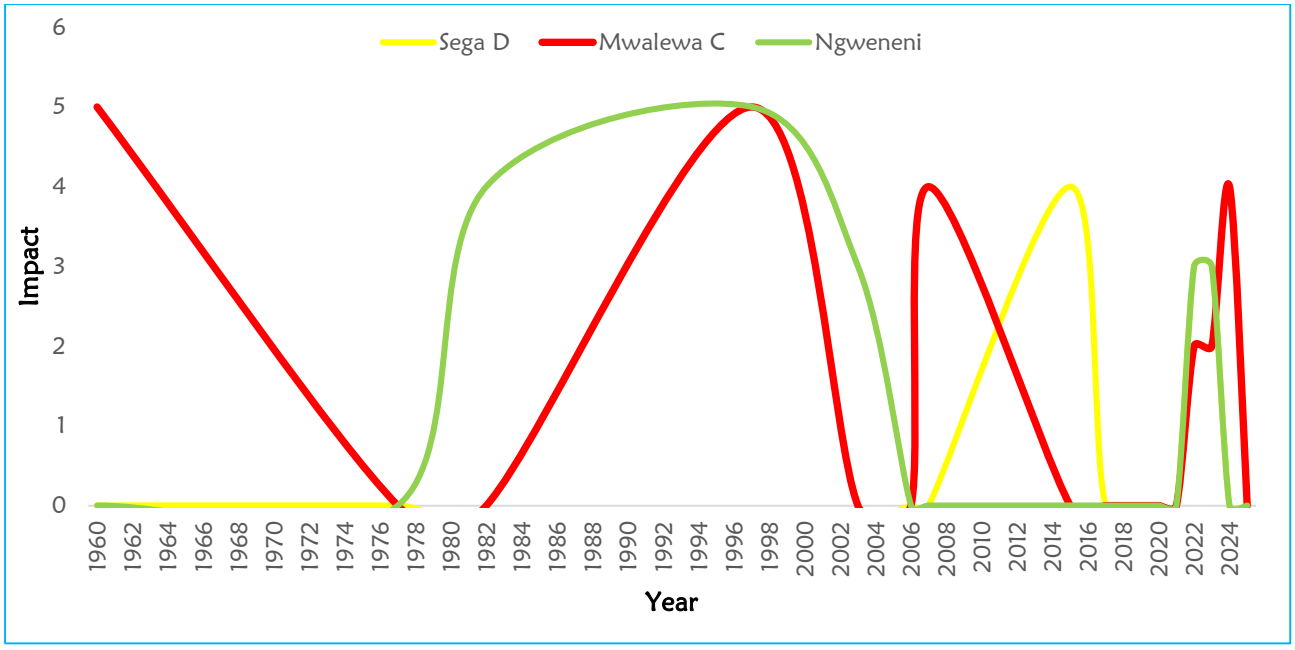

Figure 13: Floods prevalence historical trend (Source: PCVA,2025)

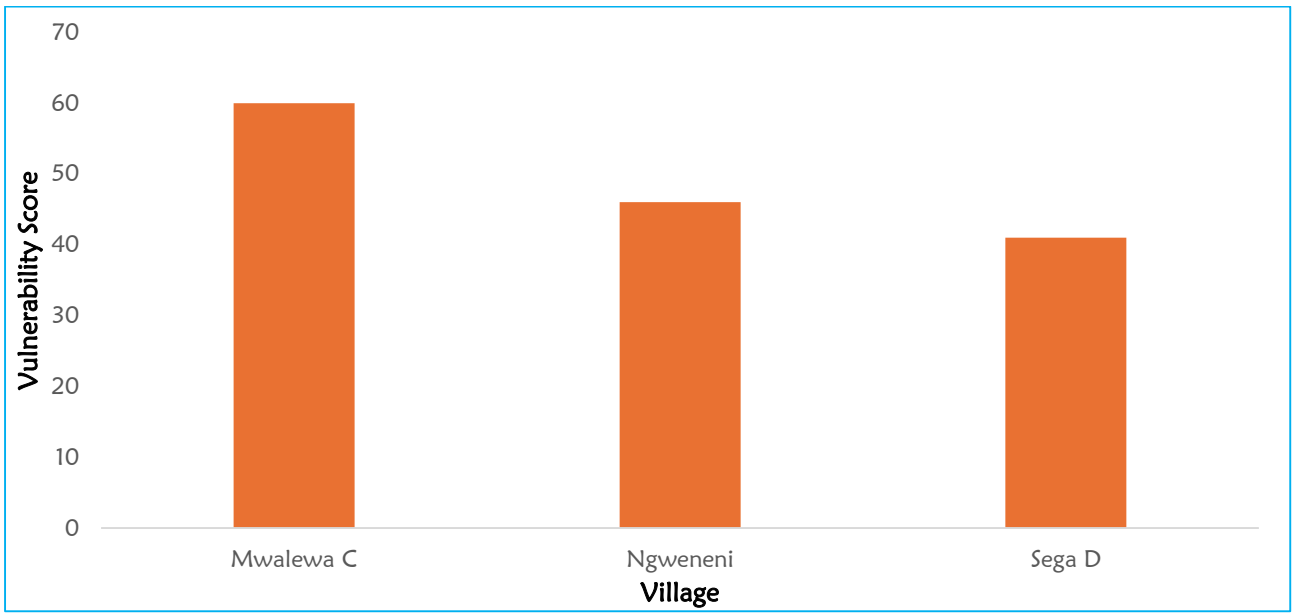

Figure 14: Floods vulnerability score in Vanga Ward (Source: PCVA,2025)

3.4.2.1 Exposure to Floods

Exposure to floods in Vanga Ward is extensive and spatially differentiated, affecting terrestrial, coastal, and riverine ecosystems as well as densely populated human settlements. The most flood-exposed terrestrial areas include Perani, Sega, Vanga, and Kiwegu units, where low-lying topography and compacted soils exacerbate surface runoff during heavy rainfall events. In these areas, recurrent inundation leads to the destruction of croplands, loss of topsoil, and erosion of access roads, thereby impeding mobility and isolating communities during peak rainfall seasons. The coastal stretch encompassing Vanga, Jimbo, Kiwegu, and Jasini is equally vulnerable, with sea-level rise and storm surges compounding the effects of fluvial flooding. This dual exposure often results in saltwater intrusion into freshwater wells and farmland, rendering soils infertile and compromising food production. Along the River Umba, extensive riverine farmlands are periodically submerged as the river overtops its banks during intense rainfall, washing away crops and displacing farming households that depend on seasonal cultivation. Settlements such

as Botola, the Lunga-Lunga border area, Mwalewa, and parts of Vanga village experience frequent flooding due to their proximity to drainage channels and inadequate stormwater infrastructure, leading to recurrent losses of property and temporary displacement of residents. These multiple zones of exposure illustrate the complex interplay between hydrological processes, land-use practices, and settlement expansion in heightening flood risks across Vanga Ward, particularly for poorer households and communities dependent on land and coastal resources for their livelihoods.

3.4.2.2 Socio-Demographic Vulnerability to Floods

Socio-demographic vulnerability to floods in Vanga Ward is shaped by gender, age, physical ability, livelihood type, and socio-economic status, all of which influence how different groups experience and recover from flood events. Women, particularly those heading households or engaged in informal trade, face heightened vulnerability due to their domestic responsibilities and limited access to productive assets. During flood periods, they bear the additional burden of caring for sick family members, collecting safe water from distant sources after wells become contaminated, and managing food shortages when access to markets and farmlands is disrupted. Children are similarly at high risk owing to exposure to waterborne diseases such as cholera and diarrhoea, unsafe play in flooded environments, and interruption of schooling when learning institutions are submerged or used as temporary shelters. Elderly persons and persons abled differently (PAD) face unique challenges related to mobility, evacuation, and access to medical services, making them more dependent on others for survival during emergencies.

Economically, low-income households, especially those residing in informal settlements near flood-prone zones like Mwalewa, Botola, and along the River Umba floodplain, are least able to recover from flood-induced losses due to limited savings and inadequate access to insurance or credit facilities. Their housing structures, often made of mud walls and thatched roofs, are easily destroyed, forcing repeated rebuilding that deepens poverty cycles. Youth engaged in casual labour, fishing, or small-scale seaweed farming also experience recurring income shocks as floods disrupt marine and agricultural production, further undermining household resilience. Conversely, a small segment of households with diversified income sources, such as salaried employment or established trade, demonstrate moderate coping ability and quicker recovery after flood events.

3.4.3 Pests and Diseases

Pests and diseases have emerged as a recurrent and increasingly severe climate-induced hazard in Vanga Ward, with clear linkages to changing temperature, humidity, and rainfall regimes. Historical analysis between 1980 and 2025 shows that pest and disease outbreaks occur approximately once every six years, with notable peaks recorded in 1980, 1995, 1997, 2003, 2017, 2021, and 2025 (Figure 16). The frequency and distribution of these outbreaks have intensified over time due to erratic rainfall patterns, warmer temperatures, and prolonged dry spells that weaken crop and livestock resistance. The Participatory Climate Vulnerability Assessment (PCVA) findings indicate that Ngweneni is the most affected area (vulnerability score 52), followed by Mwalewa C (36) and Mikoroshoni (35)(Figure 17). The persistence of these hazards has been particularly evident in Mikoroshoni, where repeated infestations of white caterpillars (locally known *as jongoo nyeupe*), locusts, and fall armyworms have devastated crops between 2017 and 2025, leading to widespread food insecurity and income loss.

Natural ecosystems and key livelihood resources across the ward have been equally vulnerable. The Gonja and Kaya Sega forests, which provide essential ecosystem services such as soil stabilization and microclimate regulation, have increasingly experienced pest-related leaf defoliation and fungal infections that impair forest regeneration. Mangrove forests along the Vanga and Kiwegu coasts are also susceptible to fungal rot and parasitic infestations during

extended periods of high temperature and humidity, leading to tree mortality and reduced fish nursery habitats. Similarly, seaweed farms, a vital source of income for coastal women and youth, have suffered recurrent disease outbreaks, including ice-ice disease caused by environmental stress and bacterial infections, resulting in large-scale productivity declines.

At the community level, exposure to pests and diseases is shaped by socio-economic and gender dynamics. Smallholder farmers and seaweed cultivators, who rely heavily on climate-sensitive resources, are the most exposed, as pest infestations and livestock diseases directly undermine their livelihoods. Women, who dominate seaweed farming and subsistence agriculture, face disproportionate impacts due to reduced harvests and increased labour burdens associated with replanting or pest management. Pastoral households, especially in Mwalewa C and Ngweneni, are affected by recurring livestock diseases such as foot-and-mouth and tick-borne infections, which intensify during wetter years following prolonged droughts.

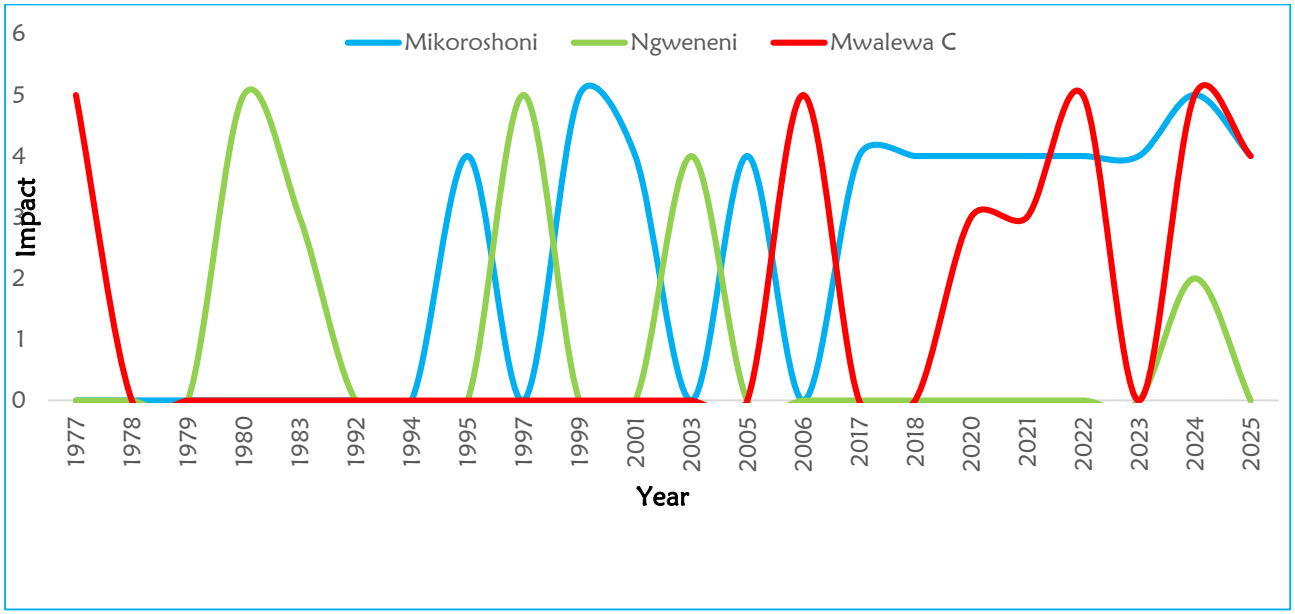

Figure 15: Pests and diseases prevalence historical trend (Source: PCVA,2025)

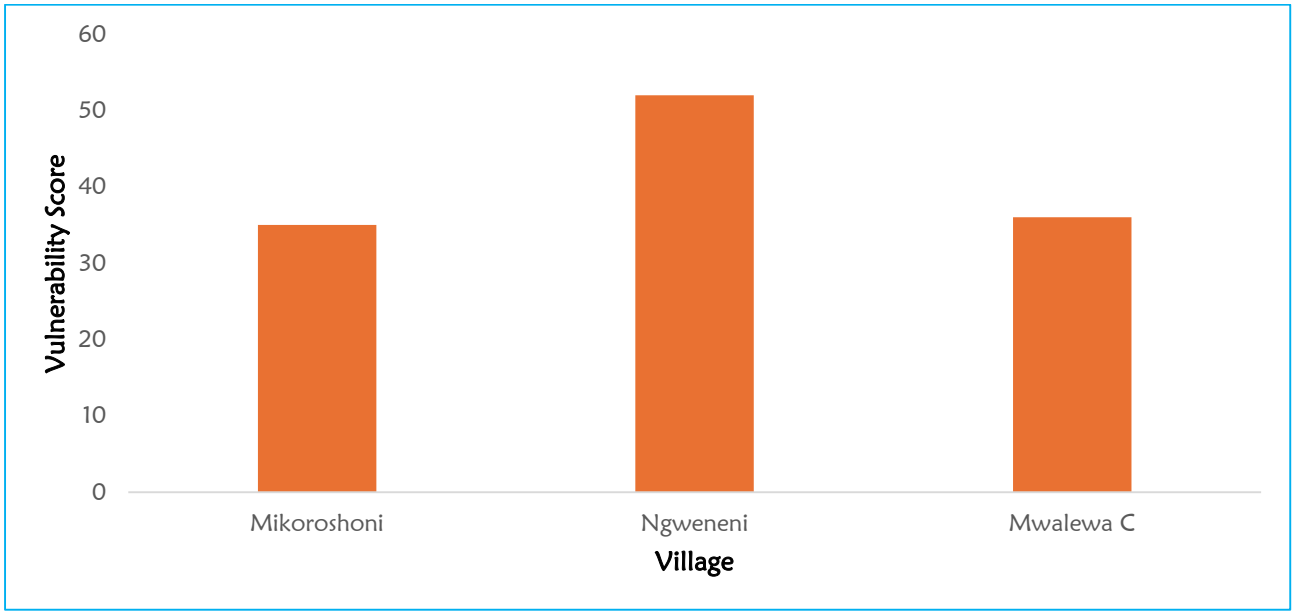

Figure 16: Pests and diseases vulnerability score in Vanga Ward (Source: PCVA,2025)

3.4.3.1 Exposure to Pests and Diseases

Exposure to pests and diseases in Vanga Ward is widespread, affecting all villages and livelihood systems across the terrestrial and coastal environments. The entire ward faces recurrent infestations and outbreaks that impact crops, livestock, and human health due to changing climatic conditions, particularly fluctuations in temperature, rainfall, and humidity. Periodic pest invasions such as locusts, armyworms, and the locally identified jongoo nyeupe (white caterpillar) have devastated farmlands in Ngweneni, Mwalewa C, and Mikoroshoni, while livestock diseases such as foot-and-mouth and tick-borne infections have increased in frequency following alternating drought and wet seasons. This exposure is heightened by the ward's dependence on rain-fed agriculture, open grazing systems, and the proximity of settlements to degraded forest and wetland habitats that harbour pest breeding sites.

3.4.3.2 Socio-Demographic Vulnerability to Pests and Diseases

Socio-demographic vulnerability to pests and diseases in Vanga Ward is deeply influenced by livelihood dependence, gender roles, age, and access to adaptive resources. The most affected groups are smallholder farmers, pastoralists, and coastal seaweed cultivators, whose livelihoods are directly dependent on climate-sensitive natural resources. Women, who form the majority of these groups, are particularly vulnerable as they manage subsistence farms, tend small livestock, and lead seaweed farming enterprises that are highly susceptible to pest and disease outbreaks. When infestations such as locusts, armyworms, and jongoo nyeupe destroy crops, or when seaweed diseases reduce harvests, women's income and household food security are severely compromised. Their adaptive capacity is further constrained by limited access to pest control technologies, extension services, and credit facilities, forcing them to rely on traditional coping mechanisms that are often ineffective under escalating climate pressures.

Youth face heightened vulnerability due to their reliance on casual labour and low-paying agricultural or marine-based livelihoods that are disrupted during pest and disease outbreaks. The loss of these income sources leads to increased unemployment and migration pressures. Elderly persons and persons abled differently (PAD) are also at high risk, as physical limitations restrict their ability to engage in pest management activities or relocate when infestations and disease outbreaks intensify. In Ngweneni and Mwalewa C, where exposure levels are highest, livestock diseases such as foot-and-mouth and tick-borne infections repeatedly erode household assets, disproportionately affecting poorer families who lack veterinary support or the capacity to restock.

From a health perspective, children and elderly individuals are more susceptible to secondary human disease outbreaks, such as malaria, dengue fever, and chikungunya illnesses, that often accompany climatic shifts and pest proliferation. In contrast, a minority of better-off households, particularly those with diversified income sources or access to agricultural extension services, exhibit moderate coping capacity through the use of improved seeds, pesticides, or seaweed treatment practices. However, the majority of vulnerable groups remain declining, with repeated losses eroding resilience over time. Overall, socio-demographic vulnerability to pests and diseases in Vanga Ward underscores entrenched gender and economic inequalities, emphasizing the need for inclusive adaptation measures that enhance women's access to information, promote climate-smart livelihoods, and strengthen community-based pest and disease surveillance systems.

3.4.4 Environmental Degradation

Environmental degradation in Vanga Ward has evolved into a persistent and multidimensional climate-related hazard, driven by unsustainable resource extraction, deforestation, and land-use change. Analysis of historical trends between 1980 and 2025 reveals that major degradation

episodes have occurred approximately once every eight years, with notable peaks in 1997 and 2024 corresponding to periods of rapid population growth, expansion of settlements, and intensified agricultural encroachment, particularly around Mikoroshoni Village (Figure 18). Over time, the frequency and severity of degradation have increased, transforming once localized disturbances into extensive ecosystem decline affecting forests, riverbanks, wetlands, and coastal zones.

The Gonja and Kaya Sega forests are among the most severely affected terrestrial ecosystems, having suffered extensive tree felling for fuelwood, charcoal, and construction materials. The depletion of forest cover has disrupted local hydrological cycles, reduced infiltration rates, and accelerated surface runoff, thereby heightening flood and drought susceptibility. Similarly, the mangrove forests in Vanga and Kiwegu have undergone significant degradation due to unregulated harvesting for fish smoking, boat construction, and timber trade. This has weakened their ecological function in shoreline stabilization, carbon sequestration, and provision of nursery grounds for fish and crustaceans. The riparian zone of the River Umba, once rich in vegetation and biodiversity, has been severely degraded by sand mining and agricultural expansion, leading to channel instability, increased sedimentation, and reduced water quality.

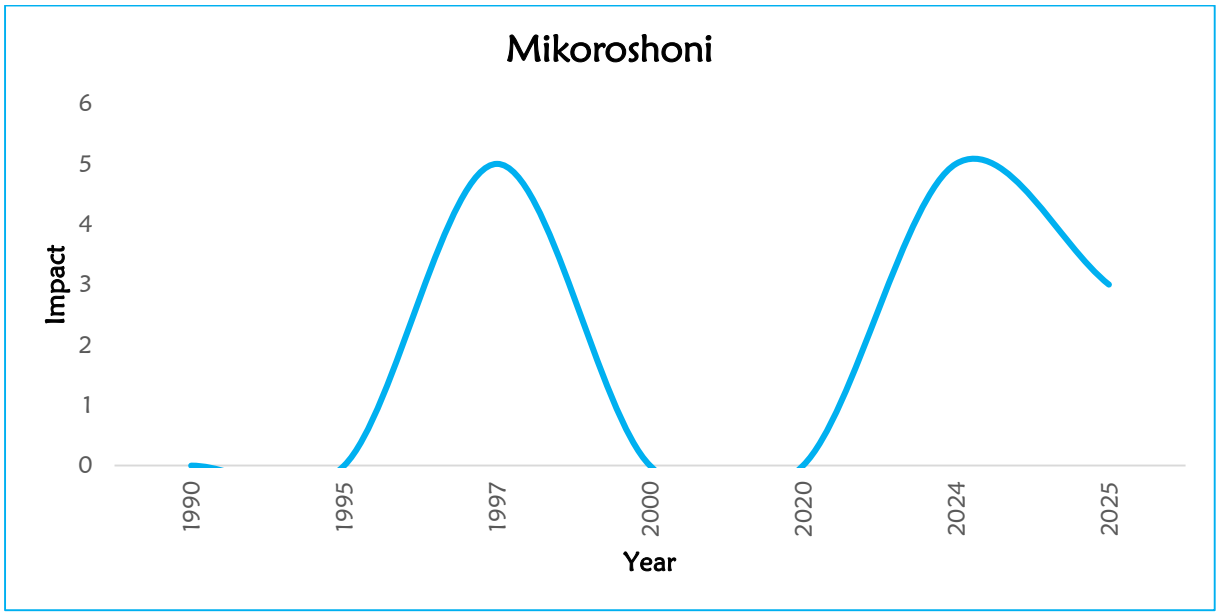

Figure 17: Environmental Degradation prevalence historical trend (Source: PCVA,2025)

3.4.4.1 Exposure to Environmental Degradation

Exposure to environmental degradation in Vanga Ward is widespread, encompassing both terrestrial and coastal ecosystems as well as the communities that depend on them for their livelihoods. The most affected areas include the Gonja and Kaya Sega forests, the mangrove belts of Vanga and Kiwegu, and the riparian zones of the River Umba, where unregulated human activities have significantly altered natural ecological functions. In the upland and inland sections, clearing of vegetation for farming, fuelwood, and settlement expansion has exposed vast tracts of land to erosion, reduced soil fertility, and disrupted local hydrological cycles. The degradation of Gonja and Kaya Sega forests has not only led to biodiversity loss but also increased the vulnerability of surrounding communities to floods and droughts due to diminished infiltration and groundwater recharge. Along the coast, the mangrove forests face ongoing exposure from overharvesting for construction and fish smoking, compounded by sea-

level rise and saline intrusion, which further weaken mangrove regeneration and reduce shoreline protection.

The River Umba riparian corridor is particularly exposed due to rampant sand mining and agricultural encroachment, which have destabilized riverbanks and increased sediment loads downstream. As a result, water turbidity has risen, negatively affecting aquatic life and irrigation water quality. Communities in Mikoroshoni, Kiwegu, and Vanga villages are at the highest exposure levels, as they rely directly on these degraded ecosystems for food, water, fuel, and income. Women and youth, who depend heavily on forest products, mangrove resources, and small-scale farming, are disproportionately affected; they must now travel longer distances to access essential natural resources, increasing their physical and economic vulnerability. In contrast, a few better-off households with diversified income sources are moderately exposed, as they rely less on natural ecosystems for daily sustenance.

3.4.4.2 Socio-Demographic Vulnerability to Environmental Degradation

Socio-demographic vulnerability to environmental degradation in Vanga Ward is strongly influenced by gender roles, livelihood dependence, income levels, and access to natural resources. The most affected groups are women, smallholder farmers, artisanal fishers, and youth, who rely directly on forests, mangroves, and riverine ecosystems for subsistence and income generation. Women, in particular, bear the greatest burden as they depend on forest and mangrove resources for fuelwood, herbs, wild fruits, and materials for fish smoking and thatching. With increasing deforestation in the Gonja and Kaya Sega forests and depletion of mangroves in Vanga and Kiwegu, women are now forced to walk long distances, often exceeding 5 kilometres daily, in search of firewood and other household necessities. This not only increases their exposure to physical exhaustion and security risks but also reduces time available for childcare, education, and economic activities, thereby deepening gender inequality.

Smallholder farmers are equally vulnerable due to declining soil fertility, loss of vegetation cover, and increasing erosion in upland areas. The degradation of forest ecosystems has disrupted natural hydrological cycles, leading to lower crop yields and reduced groundwater recharge, which in turn undermines food security and household resilience. Fisherfolk and seaweed farmers along the coast experience heightened vulnerability as mangrove degradation reduces fish breeding habitats and exposes the shoreline to erosion, affecting marine productivity and livelihoods. Youth, who often depend on casual labour in charcoal burning, mangrove harvesting, or sand mining, face unsustainable and insecure income sources that perpetuate poverty and accelerate resource depletion. Elderly persons and persons abled differently (PAD) are further marginalized, as the decline in nearby natural resources limits their access to firewood, clean water, and traditional medicinal plants.

Spatially, the highest socio-demographic vulnerability is observed in Mikoroshoni, Vanga, and Kiwegu villages, where environmental degradation has been most pronounced and alternative livelihoods are limited. Households in these areas exhibit declining adaptive capacity, trapped in a cycle where poverty drives further resource extraction, and degradation deepens poverty. Only a few moderately resilient households, typically those with diversified incomes from trade, employment, or remittances, show some improvement through reduced dependence on natural ecosystems. Overall, environmental degradation in Vanga Ward has deepened existing social inequities, with women, the poor, and resource-dependent groups bearing the greatest vulnerability. This underscores the urgent need for gender-responsive restoration programs, sustainable livelihood diversification, and community-led ecosystem management to rebuild both ecological and social resilience.

3.4.5 Strong Winds

Strong winds constitute a recurrent but localized climatic hazard in Vanga Ward, typically occurring once every ten years and associated with shifting monsoon patterns and convective storms along the coast. Historical records between 1980 and 2025 indicate major windstorm events in 2004 and 2024, both of which caused widespread structural damage and ecological disturbance, particularly in Sega D and parts of Kiwegu (Figure 19). Although less frequent than droughts and floods, the intensity and destructive potential of strong winds have increased in recent decades due to deforestation, loss of vegetation buffers, and reduced mangrove cover, which historically served as natural windbreaks.

Meteorological data from the Kenya Meteorological Department (KMD, 2024) classify wind speeds averaging 8–12 metres per second (m/s) along the South Coast (Kwale–Vanga zone) as high winds. These speeds, often recorded during the Southeast Monsoon (Kuzi) between May and September, are sufficient to uproot shallow-rooted trees, damage weak structures, and generate turbulent sea conditions. Such wind intensities are considered hazardous for artisanal fishers and coastal settlements lacking sufficient vegetative cover or infrastructure reinforcement (KMD, 2024).

The effects of strong winds are particularly severe across key natural and livelihood assets. In Gonja and Kaya Sega forests, intense gusts have uprooted mature trees and shattered canopy structures, disrupting regeneration cycles and exposing soils to erosion. Similarly, mangrove forests in Vanga, Kiwegu, and Jimbo face recurring damage as strong winds uproot seedlings and mature stands, reducing their shoreline stabilization capacity and increasing vulnerability to storm surges. Similarly, seaweed farms, a critical livelihood source for coastal women and youth, are repeatedly damaged as strong waves and currents generated by high winds tear off seaweed lines, displacing entire cultivation setups and resulting in significant income losses.

At the community level, Sega D and Kiwegu residents are the most exposed, given their proximity to open coastal and upland areas with minimal vegetative cover. Strong winds frequently damage roofs of schools, health centres, and residential houses, particularly those constructed with lightweight materials such as iron sheets and timber. Women and children face increased vulnerability as domestic spaces and learning facilities are destroyed, forcing temporary relocation and disruption of social services. Fishers operating in nearshore waters are also at heightened risk during windstorms, as strong gusts endanger small fishing vessels and limit access to marine resources. Over time, the frequency and spatial extent of wind-related hazards have expanded, driven by declining vegetation cover and ecosystem degradation.

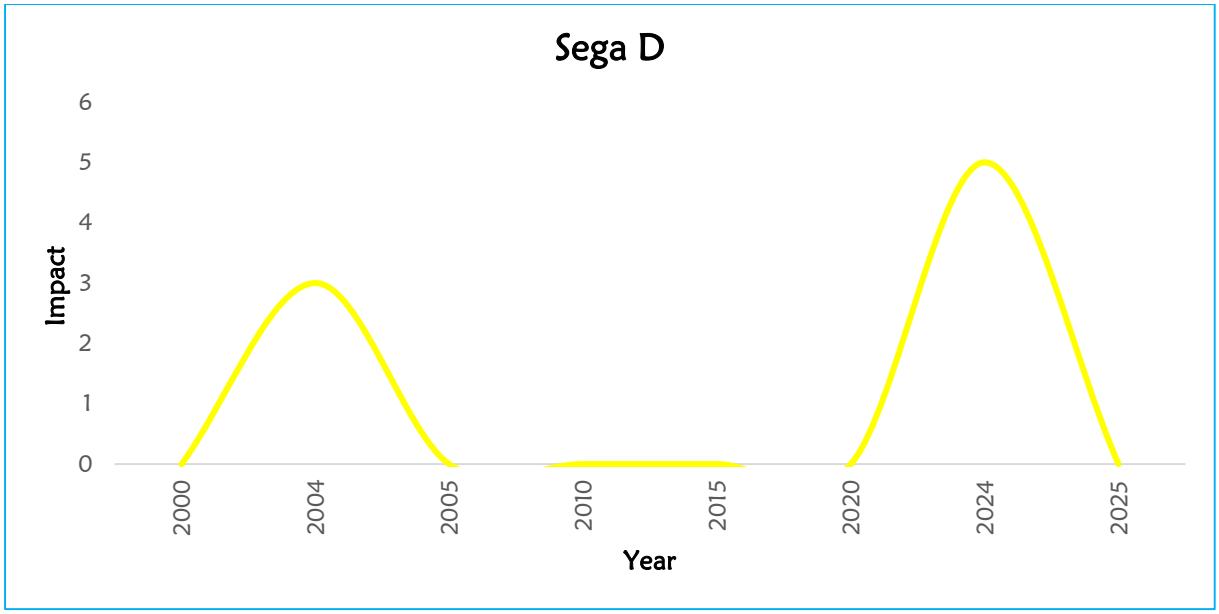

Figure 18: Strong winds prevalence historical trend (Source: PCVA,2025)

3.4.5.1 Exposure to Strong Winds

Exposure to strong winds in Vanga Ward is concentrated mainly in Sega D and Kiwegu, where open landscapes and proximity to the coastline heighten vulnerability to windstorms that occur approximately once every ten years. These winds inflict damage on both ecosystems and community assets, uprooting trees in the Gonja and Kaya Sega forests, stripping vegetation cover, and toppling mangrove stands in Vanga and Kiwegu that once acted as natural barriers. Seaweed farms along the coast are highly exposed, as strong gusts and turbulent waves repeatedly tear cultivation lines and reduce harvest yields. At the community level, settlements in exposed zones experience frequent roof damage, destruction of weak structures, and disruption of essential services such as schools and health centres. Women and youth engaged in seaweed farming, as well as fishers operating in nearshore waters, are the most exposed groups, facing direct livelihood and safety risks whenever strong winds occur.

3.4.5.2 Socio-Demographic Vulnerability to Strong Winds

Socio-demographic vulnerability to strong winds in Vanga Ward varies across gender, age, and livelihood groups, with exposure and sensitivity determined largely by settlement location, housing quality, and economic activity. Women and youth, particularly those engaged in seaweed farming and small-scale trade along the coast, are among the most affected, as strong winds frequently destroy seaweed drying racks, tear cultivation lines, and interrupt marine-based income sources. The financial losses from these disruptions disproportionately impact women-headed households, which often lack savings or access to credit for recovery. Fishers, mainly men operating small non-motorized boats, face high physical risk during strong wind events, as turbulent seas and rough currents make fishing unsafe and damage gear, leading to income instability.

Children and elderly persons are indirectly vulnerable due to the destruction of schools, health centres, and homes, which compromises access to education, shelter, and medical care. The risk is exacerbated in Sega D and Kiwegu, where many houses are built from lightweight materials such as iron sheets and timber that are easily damaged during windstorms. Persons abled differently (PAD) and low-income families face heightened vulnerability, as limited mobility and economic means constrain their ability to reinforce structures or relocate to safer areas. Over time, recurring wind damage has eroded the adaptive capacity of these groups, leading

to cycles of repair, loss, and economic strain. Overall, socio-demographic vulnerability to strong winds in Vanga Ward underscores the intersection between poverty, gender inequality, and environmental exposure, calling for targeted interventions such as resilient housing designs, livelihood diversification, and early warning systems that address the specific needs of the most at-risk populations.

3.4.6 Human–Wildlife Conflicts (HWCs)

Human–wildlife conflicts (HWC) have increasingly become a localized yet significant climate-related hazard in Vanga Ward, particularly in areas bordering degraded forest ecosystems and wildlife corridors. Historically rare, these conflicts now occur approximately once every five to six years, with a notable surge in 2025 when several incidents were reported in Mwalewa C Village, adjacent to the Gonja Forest. The rising frequency of HWC events is closely linked to the degradation of natural habitats, driven by deforestation, agricultural expansion, and changing rainfall patterns that reduce water and forage availability within forest ecosystems. As a result, wildlife species such as elephants, baboons, and monkeys increasingly move beyond forest boundaries into farmlands and settlements, causing damage to crops, livestock, and property.

The degradation of the Gonja and Kaya Sega forests has intensified exposure to HWC by diminishing the natural barriers and food sources that once confined wildlife within these ecosystems. Reduced forest canopy cover and shrinking water sources have altered wildlife movement patterns, pushing animals into nearby agricultural zones. Similarly, the ecological stress on mangrove forests and riverine vegetation has indirectly increased human–wildlife interactions as communities extend their farming activities and settlements into marginal areas after losing productive land to wildlife damage. In Mwalewa C, vulnerability analysis recorded an overall score of 36, with crop farming (score 5) and livestock keeping (score 3) being the most affected livelihood sectors. Elephants trample farmlands, baboons and monkeys raid crops, and in some cases, livestock such as goats and poultry fall prey to opportunistic wild animals.

From a socio-demographic perspective, farmers living along the Gonja Forest edge and pastoralists in open grazing areas are the most exposed groups. Women and children, who spend long hours on farms and at water collection points, face heightened physical danger from wildlife encounters, while widows, elderly persons, and persons abled differently (PAD) are particularly vulnerable due to limited mobility and inability to construct deterrent fences or invest in alternative livelihoods. The destruction of crops and livestock not only threatens household food security but also forces families to depend on purchased food or relief support, thereby deepening poverty levels. Over time, HWC incidents have expanded in both frequency and severity, reflecting a broader ecological imbalance as wildlife habitats shrink and human settlements expand.

3.4.6.1 Exposure to Human–Wildlife Conflicts

Exposure to human–wildlife conflicts in Vanga Ward is primarily concentrated in communities and ecosystems located near forested and degraded wildlife habitats, particularly around Mwalewa C Village adjacent to Gonja Forest and the surrounding landscapes of Sega and Kiwegu. These areas experience the highest interaction between humans and wildlife due to diminishing vegetation cover, shrinking water sources, and expansion of farmlands into forest buffer zones. The degradation of Gonja and Kaya Sega forests has significantly reduced natural barriers and food availability for wildlife, driving species such as elephants, baboons, and monkeys to forage in nearby farms. This exposes local farmers to frequent crop raids, destruction of fruit trees, and trampling of young seedlings, especially during dry seasons when natural forage is scarce.

The exposure extends to livestock keepers who graze animals near forest fringes and along the River Umba, where water availability attracts both domestic herds and wildlife, increasing the likelihood of predation and competition. The mangrove and coastal zones are indirectly affected, as households displaced by wildlife damage relocate their agricultural and settlement activities toward these fragile areas, heightening pressure on mangrove resources and seaweed farming sites. Women and children, who spend extended hours on farms, in forest peripheries, or collecting water and firewood, are the most exposed to physical danger from wildlife encounters, while widows, the elderly, and persons abled differently (PAD) face heightened exposure due to limited mobility and inability to erect protective fences or respond quickly to threats.

3.4.6.2 Socio-Demographic Vulnerability to Human–Wildlife Conflicts (HWCs)

Socio-demographic vulnerability to human–wildlife conflicts in Vanga Ward is shaped by livelihood type, gender, age, and proximity to wildlife habitats, with the highest vulnerability observed among resource-dependent and low-income households. Smallholder farmers and pastoralists residing near Gonja Forest and along forest edges in Mwalewa C, Sega, and Kiwegu are the most affected, as their livelihoods rely heavily on land and water resources that overlap with wildlife ranges. Frequent crop raids by elephants, monkeys, and baboons, and predation of goats and poultry, undermine household food security and income stability. These losses are especially devastating for poor households that lack access to compensation mechanisms or alternative livelihood options.

Women are disproportionately vulnerable because they perform daily activities that heighten their exposure, including farming, firewood collection, and water fetching near forested areas. The constant fear of wildlife encounters limits their mobility and reduces productivity in agriculture and trade. Children also face high vulnerability as they often accompany parents to farms or water sources, exposing them to physical danger. Widows, elderly persons, and persons abled differently (PAD) experience compounded vulnerability due to restricted mobility, limited ability to construct deterrent barriers, and dependence on external assistance for protection and recovery after incidents.

The economic ripple effects of HWCs also intensify vulnerability across the broader community. Destruction of crops and livestock forces households to purchase food, depend on relief aid, or migrate temporarily, disrupting education, social cohesion, and local markets. Over time, the frequency and severity of HWCs have deepened social inequalities, as wealthier households with stronger structures, fences, or alternative income sources recover faster, while poorer families remain trapped in cycles of loss and indebtedness. Overall, socio-demographic vulnerability to HWCs in Vanga Ward highlights the intersection of environmental degradation, poverty, and gender inequality, emphasizing the need for inclusive wildlife management strategies, gender-responsive compensation schemes, and community-led mitigation measures that protect the most at-risk groups while promoting coexistence between people and wildlife.

3.5 Classification of Exposure and Resilience through Livelihoods

The ward is characterised by livelihoods that are highly exposed to environmental, economic and institutional shocks, while its capacity for resilience remains constrained by limited income diversity, low margins and persistent poverty. The dominant cash-income sources are informal and closely tied to marine/coastal and transport services with modest supplementary incomes from micro-enterprises, small-scale agriculture and livestock or poultry. At the same time, county data indicate that Kwale's poverty headcount has been around 47 per cent, higher than the national average. These factors combine to yield a system with narrow buffers and high vulnerability, thereby demanding careful classification of exposure and resilience through the livelihood lens.

3.5.1 Livelihood Structure and Monthly Incomes

Table 2 below presents differentiated typical net monthly earnings (KES) per actively engaged adult in each of the four villages (Mikoroshoni, Mwalewa C, Ngweneni, Sega D). The data illustrates that income levels are modest and vary across the four villages. Motorbike operations and fishing form the upper-income tier, with monthly returns ranging from KES 27,000 to 32,000 and KES 8,500 to KES 11,500 respectively

The high returns from motorbike operations are largely attributed to their central role in daily household logistics and community mobility. Most residents rely on motorbikes to transport children to and from school and to access markets, health facilities, and social amenities. On average, households spend about KES 200 per day on school transport alone, creating a steady and reliable demand for motorbike services. This dependence has positioned motorbike operators as critical economic actors within the ward, contributing to consistent income flows despite periodic shocks such as fluctuating fuel prices, poor road infrastructure, and mechanical breakdowns.

Fishing, while also a key livelihood source, particularly for men along the coastal and estuarine areas, remains exposed to multiple vulnerabilities, including declining fish stocks, unpredictable weather patterns, and restrictions on access to fishing zones. These risks often translate to unstable earnings, especially during closed seasons or periods of rough seas.

In contrast, livelihoods such as small-scale trade, handicrafts, poultry keeping, and subsistence agriculture yield significantly lower monthly incomes, typically ranging between KES 1,000 and KES 5,000. These activities are predominantly undertaken by women and youth and are characterized by limited capital investment, small market reach, and high sensitivity to seasonal variations. Consequently, households engaged primarily in these sectors exhibit low financial resilience, with minimal capacity for savings or reinvestment.

Table 2: Monthly earnings by livelihood and village (Source: PCVA, 2025)

Livelihood	Income (KES)			
	Mikoroshoni	Mwalewa C	Ngweneni	Sega D
Motorbike operations	28,000	32,000	27,000	30,000
Fishing	11,500	9,000	8,500	10,500
Small scale business	4,800	4,500	5,000	4,200
Agriculture	1,300	1,600	1,800	1,200
Weaving/handicrafts (mats, baskets)	1,500	1,100	1,350	—
Tree-seedling nursery sales	3,600	4,000	3,200	—
Chamas (community savings groups)	500	450	550	400
Goat rearing	1,100	—	—	—
Livestock production	5,200	4,600	4,900	5,000

The poverty profile of the reflects the economic fragility of its livelihoods. Field data show that many households in all four villages experienced prolonged periods without adequate income to meet their basic needs over the past year. Figure 20 below presents the number of people who reported lacking money to meet critical needs such as food, housing, clothing, transport, and medication in the last 12 months.

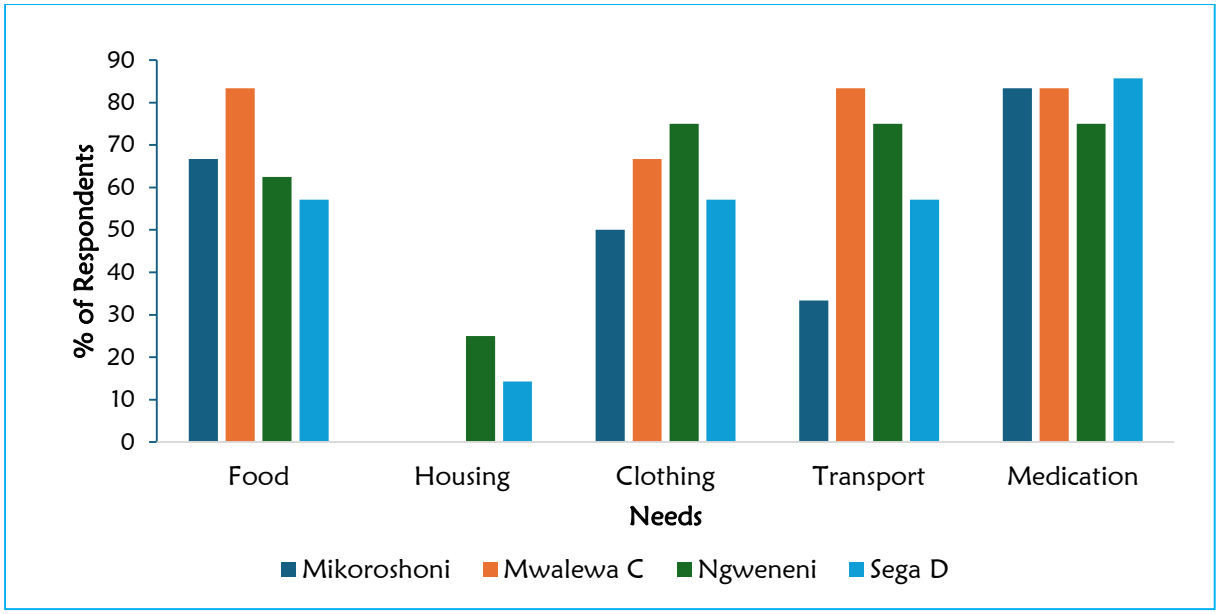

Figure 19:Respondents who lacked money to Meet Basic Needs in the Last 12 Months (Source: PCVA 2025)

This data provides evidence of widespread deprivation across the ward. The inability to afford food and medication is particularly striking, with between 63% and 86% per village unable to meet such needs. Medication deprivation, ranging from 75% to 86% per village, suggests that a significant proportion of households are unable to access healthcare, which heightens vulnerability to illness and undermines productivity. In addition, high numbers of people lacking transport and clothing reflect the broader structural poverty that limits both access to markets and participation in income-generating activities.

Poverty in Vanga Ward thus manifests as a chronic, multidimensional phenomenon rather than a temporary financial shortfall. The 2023 Kwale CIDP confirms that coastal zones such as Vanga continue to experience high poverty rates, with over 45% of households living below the county's basic-needs threshold. This economic deprivation perpetuates a cycle of low resilience: households with minimal disposable income are unable to invest in livelihood improvement, asset replacement, or climate adaptation measures. Consequently, each shock, whether climatic, health-related, or economic, deepens vulnerability, forcing negative coping strategies such as reduced meals, sale of productive assets, or child labour.

3.5.2 Vulnerabilities of Livelihood Options

All livelihood options in the ward while essential for household sustenance, expose the community to varying degrees of risk. Motorbike operations, though offering the highest cash income, are highly sensitive to fluctuations in fuel prices, spare parts costs, and weather-related road disruptions. Heavy rains and flooding frequently render rural roads impassable, reducing daily income and increasing maintenance costs. Furthermore, accident risks and lack of insurance expose operators to catastrophic losses, often plunging households back into poverty.

Fishing, the second most important livelihood, remains extremely vulnerable to environmental and regulatory pressures. Declining fish stocks due to overfishing, mangrove degradation, and illegal gear use, combined with changing ocean currents and rough sea conditions, frequently reduce catch volumes. The economic impact of such ecological stressors is compounded by limited cold-chain infrastructure and weak governance among Beach Management Units (BMUs), which undermines effective resource management. These vulnerabilities mirror the

CIDP's acknowledgment that while fisheries contribute significantly to Kwale's GDP, the sector remains under-capitalized and climate sensitive.

Micro-retail and trade activities, practiced mainly by women and youth, offer minimal financial returns and are constrained by limited capital, seasonal demand, and competition. Flooding and poor drainage often disrupt trading days, while low purchasing power within the local market restricts profitability. Rain-fed agriculture faces recurrent climate shocks, including droughts and pest infestations, leading to unstable yields. Saline soils and low soil fertility along the coast further limit crop diversity and productivity, forcing households to depend heavily on purchased food.

Auxiliary activities such as weaving, poultry, and tree-seedling production provide diversification but are equally fragile. Weaving relies on access to raw materials such as coconut fronds and mangrove reeds, whose availability fluctuates seasonally. Poultry production faces recurrent disease outbreaks and high feed costs, while seedling production is sensitive to drought, requiring regular watering and nursery maintenance. Goat rearing, though practiced only in Mikoroshoni, remains a low-scale activity constrained by limited grazing areas and veterinary services.

Underlying all these vulnerabilities is the persistent issue of poverty. The low income levels indicate that most households operate within or just above subsistence thresholds, leaving little room for investment or shock recovery. In this context, even minor external disturbances, such as a week of poor fishing weather, a spike in fuel costs, or an illness in the family can severely undermine household stability. The reliance on informal, low-profit activities without access to formal financial services amplifies this fragility.

3.5.3 Overall Classification and Implications

The cumulative evidence positions Vanga Ward within a high-exposure, low-to-moderate-resilience category. The community's economic dependence on climate-sensitive and informal livelihoods, combined with high poverty levels and gendered unpaid work burdens, restricts adaptive capacity and heightens vulnerability. Addressing these constraints requires integrated measures that improve income stability in high-risk sectors, promote diversification into higher-value chains, reduce women's unpaid workload through labour-saving interventions, and strengthen social safety nets. Expanding access to microfinance, improving infrastructure such as roads and cold storage, and investing in ecosystem-based adaptation particularly mangrove restoration and sustainable fisheries will be vital in shifting the ward's resilience profile from reactive coping to proactive adaptation.

3.6 Analyzing causes of vulnerability.

3.6.1 Classification of Severity of the Vulnerabilities and Their Causes

The ward exhibits varying degrees of vulnerability across its four sampled villages, Mikoroshoni, Sega D, Ngweneni, and Mwalewa C, largely determined by their ecological location, livelihood dependence, and adaptive capacity. The classification was informed by participatory scoring and spatial analysis. Mikoroshoni and Sega D demonstrated high vulnerability due to their proximity to the coastline, dependence on marine resources such as seaweed and mangroves, and exposure to saline intrusion and floods. Ngweneni and Mwalewa C were classified as moderately vulnerable, where drought, pest infestation, and limited water access threaten agricultural productivity.

The severity of vulnerability is compounded by limited livelihood diversification, chronic poverty, and inadequate institutional support. Communities relying on mangroves and seaweed are more exposed to climatic fluctuations, while those depending on agriculture face

recurrent crop failures. Women experience disproportionate vulnerability due to their engagement in low-return economic activities, restricted access to credit and training, and the intensification of unpaid domestic work during climate shocks. This classification underscores the multidimensional nature of vulnerability, where physical exposure, socio-economic status, and gender intersect to determine the overall risk profile.

3.6.2 Identification of Unsafe Conditions

Unsafe conditions in the ward refer to the immediate settings and circumstances that heighten exposure to climate-related hazards. Settlements located along the tidal zones in Mikoroshoni and Sega D face recurrent flooding and saltwater intrusion, resulting in frequent loss of assets and disruption of livelihoods. Inadequate drainage systems and poor solid waste management worsen flood impacts, while deforested areas around the mangrove belt expose the coastline to erosion and storm surges.

From a gender perspective, unsafe conditions are further reflected in the burden placed on women who must travel longer distances to collect water and firewood during dry seasons, often through degraded landscapes that pose physical risks. Limited access to climate-resilient infrastructure such as raised storage facilities, safe fish landing sites, and strong wind-resistant shelters intensifies vulnerability for both men and women, but particularly for households headed by widows or persons abled differently.

3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability

Dynamic pressures represent the intermediate processes through which underlying socio-economic and institutional weaknesses are translated into direct, unsafe conditions that heighten vulnerability. In Vanga Ward, these pressures manifest through rapid population growth, unregulated land-use change, overexploitation of natural resources, and weak enforcement of environmental by-laws, all of which interact to degrade ecosystems and undermine resilience.

The Gonja Forest Block, one of the ward's key ecological assets, exhibited moderate but widespread degradation, with approximately 58.82 hectares affected. The most severe degradation occurred adjacent to Mwalewa Village, where a large contiguous area of forest has been disturbed, while smaller and fragmented patches covering 11.76 hectares (3.98%) were identified within Sega Sub-location. The highest concentration of degradation was recorded in Jego Sub-location, affecting 47.06 hectares (23.54%) of the total forest area. The primary drivers of this decline are extensive livestock grazing, which impedes natural regeneration through trampling and browsing of seedlings, and unsustainable tree harvesting for construction materials and fuelwood. These pressures indicate persistent livelihood-related dependencies on forest resources and inadequate regulation of extraction activities. The degradation of the Gonja Forest Block not only reduces vegetation cover and biodiversity but also compromises its ecosystem services, such as soil stabilization, microclimate regulation, and carbon sequestration, thus amplifying community exposure to droughts, strong winds, and floods.

Beyond the forest ecosystem, increasing settlement encroachment along mangrove zones, driven by scarcity of arable land inland, has intensified competition for space and accelerated the loss of protective vegetative buffers. The decline in fish stocks and reduced seaweed productivity have also prompted households to diversify into resource-dependent coping strategies, including charcoal production and mangrove pole harvesting. These adaptive but unsustainable responses perpetuate ecological degradation, creating a feedback loop between livelihood stress and environmental decline.

From an institutional perspective, weak coordination between county departments, local community structures, and national agencies constrains effective resource governance. Enforcement of conservation by-laws remains sporadic, while early warning systems for floods and droughts are either absent or poorly operationalized, reducing preparedness and timely response. Economically, fluctuating market prices for seaweed and limited access to credit facilities continue to undermine household income stability. On the social front, the exclusion of women from decision-making forums, particularly in forest and fisheries management committees, limits the inclusivity and sustainability of adaptation initiatives.

3.6.4 Identification of Underlying Causes of Vulnerability

The underlying causes of vulnerability in Vanga Ward stem from interlinked socio-economic, environmental, and institutional deficiencies that collectively erode household resilience and adaptive capacity. Structural poverty remains pervasive, limiting access to education, technology, financial services, and diversified livelihood opportunities. According to the Participatory Climate Vulnerability Assessment (PCVA, 2025), the average household size in Vanga Ward is 7 persons per household, indicating a high dependency ratio and increased pressure on limited natural and financial resources. Larger households require greater quantities of food, water, and energy, yet often rely on subsistence livelihoods and ecosystem-dependent activities such as fishing, mangrove harvesting, and small-scale agriculture. This dependency entrenches poverty cycles and increases exposure to environmental and economic shocks.

Environmental degradation constitutes another major underlying factor, manifesting in mangrove deforestation, soil erosion, declining marine productivity, and degradation of adjacent terrestrial ecosystems such as the Gonja Forest Block. The loss of vegetative cover reduces the capacity of ecosystems to buffer against climate-induced hazards, particularly floods, droughts, and strong winds, thereby increasing community exposure and reducing long-term ecological stability.

Institutionally, weak governance frameworks and limited coordination between the County Government of Kwale, the Ward Climate Change Planning Committee (WCCPC), and relevant national agencies hinder the implementation of adaptive interventions. Resource constraints, overlapping mandates, and inadequate policy enforcement reduce the effectiveness of climate action at both ward and county levels. Furthermore, the absence of comprehensive early warning and information dissemination systems results in delayed community responses to climatic hazards, shifting the focus from preparedness to crisis management.

Gender inequality further compounds vulnerability. Patriarchal norms and customary land tenure systems restrict women's ownership and control of productive assets, constraining their participation in decision-making and their capacity to adopt climate-smart technologies or access credit facilities. Consequently, women remain overrepresented in low-return and climate-sensitive livelihood activities, limiting their economic independence and adaptive potential.

3.6.5 Prioritization of Causes

Prioritization of the identified causes was achieved through participatory ranking during the PCVA exercises, considering frequency, severity, and reversibility of impacts. The highest priority causes were poverty, environmental degradation, and weak institutional support systems. Poverty restricts adaptive investments and perpetuates reliance on climate-sensitive livelihoods, while environmental degradation reduces the ecosystem's regenerative capacity. Weak institutional mechanisms delay response and hinder the implementation of ward-level adaptation initiatives.

Medium-priority causes include gender inequality, population pressure, and lack of access to climate information, all of which significantly influence vulnerability dynamics. Addressing these priority causes requires an integrated approach, strengthening governance structures, enhancing livelihood diversification, and promoting gender-responsive adaptation strategies. Interventions such as mangrove and kaya restoration, provision of microcredit for women's enterprises, and establishment of community-based early warning systems are essential to reduce long-term vulnerability and build resilience across all villages in the ward.

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards

Gender and age significantly influence exposure, sensitivity, and adaptive capacity across households in Vanga Ward (Table 3). The Ward experiences recurrent droughts, seasonal floods, environmental degradation, strong winds, pests and diseases, and persistent human–wildlife conflict, especially baboon and crocodile attacks in villages adjacent to Gonja Forest and River Umba, where households depend on riverine access for fishing, water collection, washing, rice cultivation, and small-scale irrigation. These hazards interact with entrenched gender inequalities, socio-economic marginalisation, and livelihood dependence on climate-sensitive sectors such as artisanal fishing, mangrove harvesting, small-scale agriculture, and beachside trading (UN Women, 2022; IPCC, 2022).

Women and girls in Vanga Ward shoulder a disproportionate burden of climate impacts due to socially ascribed domestic and care tasks, limited access to productive assets, and their overrepresentation in low-income, climate-sensitive livelihood activities. Across the Ward, women perform unpaid care and household work including cooking, childcare, laundry, collecting water and firewood, beachside trading, fish sorting, and participating in mangrove harvesting or shellfish gathering. During PCVA exercises, this unpaid workload was valued at KES 2,000 per day, confirming its economic significance despite its invisibility in formal economic reporting (KNBS, 2023; UNDP, 2022). Climate hazards intensify this workload, creating "time poverty" that restricts women's participation in paid labour, adaptation planning, training, and leadership roles (ILO, 2018; UN Women, 2023).

Drought severely affects Vanga Ward by reducing freshwater availability, damaging subsistence crops, and contributing to saltwater intrusion in coastal wells and riverine systems. Women and girls walk longer distances in search of clean water and fuelwood, often under high heat, increasing fatigue, dehydration, musculoskeletal strain, and psychosocial stress (UNICEF, 2021). Small-scale agriculture, primarily managed by women, is negatively impacted, reducing household food stocks. Men engaged in fishing and farming experience declining productivity as fish distribution shifts with warming waters and drought reduces crop viability (FAO, 2022). Youth lose casual labour opportunities in farms, fishing boats, and mangrove-related activities. Elderly persons and Persons Abled Differently (PADs) experience both reduced mobility and heightened dependency during extended droughts, limiting access to food, water, and relief supplies (WHO, 2021).

Floods exacerbate vulnerabilities by damaging homes, fishing gear, mangrove stands, farms, and local infrastructure. Women's caregiving burdens intensify as they manage sick or displaced family members, salvage household items, and secure food during market disruptions. Flooded boreholes, pit latrines, and wells increase exposure to waterborne diseases, forcing women and girls to find alternative water sources at longer distances (UNICEF, 2020). Men engaged in fishing, transport, and wage labour lose income when landing sites, boats, and roads are flooded. Youth face interrupted schooling, increased idleness, and heightened exposure to negative coping pathways. Elderly persons and PADs struggle to evacuate due to mobility barriers, reliance on caregivers, and limited access to safe shelter (WHO, 2020).

Environmental degradation, including mangrove loss, deforestation in Gonja Forest, coastal erosion, and declining fisheries, directly affects household livelihoods. Mangrove depletion intensifies storm surge impacts, increases coastal flooding, and reduces availability of fuelwood and non-timber forest products (NTFPs). Women must walk longer distances to collect firewood and wild foods, reducing time for childcare and income generation. Men incur increased costs and risks as they travel farther offshore to fish, while youth lose opportunities in nature-based enterprises. Elderly persons and PADs experience heightened dependence and reduced access to degraded resource systems.

Pests and diseases, including crop pests, water-borne diseases, and livestock infections, disrupt food security and livelihoods. Women managing subsistence plots, poultry, and goats experience reduced production and higher labour burdens. Men face financial losses from declining crop yields and livestock deaths. Youth lose income from farm and fishing-related casual work, while elderly persons and PADs rely heavily on family members when food stocks decline.

Human–wildlife conflict is a significant hazard in Vanga Ward, particularly in villages bordering Gonja Forest and River Umba, where crocodiles, snakes, and occasionally baboons threaten community safety. Women and girls are at especially high risk when fetching water, washing clothes, or farming near the riverbanks. Fear of wildlife attacks constrains women's mobility, reduces participation in livelihood activities, and increases stress (UNEP, 2022). Men guarding farms and fishing along the river face direct risk of injury. Youth assisting with grazing or water collection face increased exposure to wildlife threats. Elderly persons and PADs are extremely vulnerable due to slow mobility and limited ability to escape when wildlife incursions occur.

Strong winds damage houses, fishing gear, school structures, boats, and coastal infrastructure. Women bear increased domestic workload following damage to homes and food storage facilities. Men, especially fishermen, face risks at sea due to capsizing, gear loss, and accidents. Youth are at risk from falling trees and debris, while elderly persons and PADs struggle to relocate quickly to safe shelter.

Across all hazards, gendered labour patterns, socio-economic inequalities, and age-related limitations deepen vulnerability among women, girls, youth, elderly persons, and PADs in Vanga Ward. Effective adaptation must therefore prioritise gender-responsive, age-inclusive, and locally grounded interventions that strengthen resilience across the Ward (IPCC, 2022; UN Women, 2023; UNDP, 2022).

Gender-based violence (GBV) further compounds these gendered vulnerabilities, functioning both as a climate-induced risk and as a structural barrier to resilience in Vanga Ward. Emerging global evidence demonstrates a strong association between climate stressors and rising GBV incidence, with every 1°C increase in temperature linked to a 4.7% rise in intimate partner violence (UN Spotlight Initiative, 2024). Heatwaves have been associated with a 28% increase in femicide, while climate-driven displacement heightens risks of child marriage, sexual exploitation, and trafficking (UN Spotlight Initiative, 2024). In Kenya, rising temperatures and socioeconomic stressors have coincided with escalating femicides with over 930 cases documented between 2016 and 2024, highlighting how climate stress intersects with entrenched gender inequalities (Kenya Femicide Data Study, 2024). These dynamics are most acute in fragile, resource-dependent communities such as Vanga, where drought, floods, and livelihood loss amplify household tensions, reduce economic security, and disproportionately expose women and girls to physical, psychological, and sexual violence. GBV undermines climate resilience by restricting women's mobility, decision-making autonomy, access to productive assets, and participation in ward-level adaptation processes (UN Women, 2022). As such, GBV must be integrated into anticipatory climate action, beyond response and preparedness, through early warning systems, forecast-based triggers, community surveillance, and pre-positioned support services coordinated with mandated county structures such as the County Gender Department, Social Services, child protection units, and GBV coordination mechanisms that programmes like COSME have actively engaged. Embedding GBV mitigation within resilience planning ensures that climate adaptation measures safeguard rights, safety, and justice, which are foundational to sustainable and inclusive resilience building.

Table 3: Gender/Age Vulnerability Analysis for Climate Hazards

Women	Men	Youth	Elderly	PADs
Drought				
• Female-headed households bear highest burden of water, food, and fuelwood provision • Walk longer distances to depleted water sources → fatigue, dehydration, back strain • Reduced time for childcare, rest, and income generation • Increased psychological stress due to food shortages	• Livelihood insecurity from reduced crop yields, livestock productivity, and forest resources • Increased financial pressure and stress • Reduced availability of agricultural labour opportunities	• Reduced productivity and income from casual labour losses • High idleness → increased risk of crime, drugs, and unsafe coping strategies • Higher incidence of early marriages and pregnancies among girls	• Mobility limitations restrict access to water and food • Increased susceptibility to heat stress, dehydration, and hunger • Dependence on caregivers worsens during drought periods	• High dependency for water, food, and healthcare access • Mobility constraints limit ability to queue for relief supplies • Increased risk of neglect during prolonged shortages
Floods				
• Increased caregiving burdens for sick or displaced family members • Loss of income from informal trade due to market inaccessibility	• Loss of fishing, transport, and casual labour opportunities • Psychological stress from inability to provide for families	• School closures disrupt learning and training • Loss of casual labour -increased idleness, crime, and drug use • Girls particularly vulnerable to early	• High susceptibility to waterborne diseases and cold exposure • Limited mobility restricts evacuation • Greater dependence on others for shelter,	• Severe mobility barriers during evacuation and accessing aid • High risk of injury and neglect in shelters • Limited access to emergency services

Women	Men	Youth	Elderly	PADs
• Contaminated wells force longer distances to find safe water • Elevated psychosocial stress	• Damage to tools and assets disrupts livelihoods	marriage and pregnancy	food, and medicine	
Pests and diseases				
• High exposure due to their role in managing subsistence farms, small livestock, and seaweed farming • Reduced food security and income from harvest losses • Limited access to pest-control technologies and extension services	• Income loss from crop and livestock declines • Increased stress due to reduced labour demand and food shortages • Higher pressure to diversify livelihoods	• Loss of agricultural casual labour opportunities • Increased unemployment and financial vulnerability • Greater exposure to negative coping behaviours	• Increased dependence for food and care during outbreaks • Reduced capacity to protect crops and livestock • Nutritional deficits worsen chronic illnesses	• High dependency on family for food and adaptive support • Limited access to adapted technologies and extension services • Increased vulnerability from household food shortages
Human wildlife conflicts				
• Restricted mobility due to fear of wildlife attacks • Loss of crops and livestock increases household food insecurity	• High risk of injury when guarding farms and livestock, often at night • Economic strain from repeated	• Increased risk when helping to guard farms • Reduced productivity and fewer livelihood opportunities	• Very limited capacity to respond to wildlife encounters • Increased dependence after livestock or crop losses	• Inability to flee or protect themselves during wildlife incidents • Extremely high dependence on caregivers for

Women	Men	Youth	Elderly	PADs
• Limited access to compensation processes	crop and livestock losses • Pressure to seek alternative income sources	• Higher vulnerability to migration and exploitation	• Heightened food insecurity	safety and recovery • High vulnerability during compensation or reporting processes
Environmental Degradation				
• Reduced access to firewood, mangroves, herbs, and NTFPs • Walk longer distances (5+ km) for resources leading to exhaustion • Less time for childcare and economic activities	• Declining fish stocks increase offshore travel distance and fuel costs • Higher fishing risks • Reduced forest resource availability increases livelihood strain	• Fewer forest-based livelihood options (beekeeping, nurseries, eco-tourism) • Increased involvement in illegal extraction when opportunities shrink • Rising idleness and economic vulnerability	• Mobility challenges increase dependency during resource scarcity • Physical strain from long distances to collect resources • Heightened vulnerability to malnutrition	• Limited access to degraded resource areas • Exclusion from resource-based livelihood opportunities • Increased reliance on caregivers
Strong winds				
• Increased caregiving demand after home damage • Higher risk securing household assets during storms	• High danger at sea and gear damage • Increased risk of capsizing during strong winds	• High injury risk from collapsing structures or debris • Limited ability to seek safe shelter	• Difficulty accessing safe shelter quickly • High injury and exposure risk	• Severe challenges evacuating safely • High exposure to falling debris and structural collapse

4.1 Children's Vulnerability to Climate Hazards

Children in Vanga Ward are among the most climate-vulnerable populations due to their rapid developmental needs, dependency on caregivers, and limited ability to independently anticipate or respond to environmental hazards (UNICEF, 2022; WHO, 2020). Drought, floods, environmental degradation, strong winds, pests and diseases, and human–wildlife conflicts significantly disrupt children's access to food, clean water, safety, healthcare, education, and emotional stability, with long-term effects on growth and development.

During droughts, children in Vanga Ward face heightened exposure to acute malnutrition, stunting, anaemia, and dehydration due to declining agricultural productivity, salinity in water sources, and reduced fish availability. Water scarcity forces households to rely on unsafe, contaminated, or distant water sources, exposing children to diarrhoea, typhoid, skin infections, and dehydration (UNICEF, 2022). In many households, older children, especially girls, participate in water collection, walking long distances under extreme heat, which increases fatigue, physical strain, and school absenteeism.

Floods damage homes, ECD centres, schools, fishing infrastructure, and sanitation systems, placing children at high risk of drowning, injuries, respiratory infections, diarrhoeal diseases, and exposure to contaminated floodwater (WHO, 2020). When schools are used as temporary shelters, learning continuity is interrupted, school feeding programmes are disrupted, and children experience anxiety, overcrowding, and a loss of routine. Displacement creates emotional stress, fear, and behavioural issues among younger children.

Human–wildlife conflict presents a significant hazard for children in villages bordering Gonja Forest and River Umba. Children fetching water, playing near riverbanks, grazing livestock, or accompanying caregivers face direct risk of crocodile attacks and snake bites (UNEP, 2022). The psychological impact of witnessing or experiencing wildlife-related injuries is substantial, leading to long-term fear, trauma, nightmares, and restricted mobility.

Environmental degradation, including mangrove loss, coastal erosion, deforestation, and declining fisheries, weakens household food security and income stability. Reduced fish availability undermines protein intake for children, increasing nutritional deficiencies. Strong winds damage homes and schools, exposing children to injury, falling debris, and harsh weather conditions. Repeated shelter damage disrupts sleep, safety, and emotional well-being.

Pests and diseases affecting crops, livestock, and fisheries reduce household food stocks and incomes, weakening dietary diversity and increasing susceptibility to infections. When household resources shrink, children may be withdrawn from school, miss meals, or be required to support household labour, increasing long-term vulnerability (UNFPA, 2022).

Poverty intersects with climate hazards to produce harmful coping mechanisms such as early marriage for girls, increased child labour, reduced meal frequency, or school dropout. These strategies violate children's rights and reinforce intergenerational cycles of vulnerability and deprivation (UNDP, 2022).

Protecting children in Vanga Ward requires a comprehensive, multi-sectoral adaptation approach. Strengthening child-sensitive early warning systems ensures that households, schools, and community structures receive timely and accessible hazard information. Enhancing nutrition surveillance and implementing early action protocols is essential to identify and treat malnutrition during droughts and food shortages. Climate-resilient WASH infrastructure, including elevated latrines, protected wells, rainwater harvesting systems, and water treatment

mechanisms, is vital for preventing disease outbreaks during floods and prolonged dry periods (WHO, 2020).

Strengthening education infrastructure through storm-resistant roofing, elevated classrooms, safe water supply, and resilient storage facilities helps maintain learning continuity during emergencies (UNESCO, 2021). Establishing psychosocial support programmes, child-friendly spaces, and trauma-informed services reduces emotional distress during and after climate shocks. Expanding social protection, such as school feeding programmes, cash transfers, and targeted support for vulnerable households, stabilises children's well-being when livelihoods decline (UNDP, 2022). Enhancing the capacity of teachers, community health volunteers, religious leaders, and caregivers ensures that children's safety, health, and well-being remain central during climate response and recovery.

5 Chapter 5: Adaptation Actions

5.1 Introduction

Building on the findings of the hazard and vulnerability analysis, this chapter outlines the adaptation actions designed to strengthen climate resilience in Vanga Ward. These actions were developed through participatory consultations that utilized Participatory Climate Vulnerability Assessment (PCVA) tools and impact chain analysis to ensure that proposed measures align with the ward's ecological characteristics and socio-economic context. The interventions are tailored to address both the immediate climatic stresses, such as droughts, floods, Pests and diseases, environmental degradation, human wildlife conflicts and strong winds, and the underlying structural vulnerabilities related to poverty, ecosystem degradation, and institutional capacity gaps.

The proposed measures encompass short-, medium-, and long-term strategies. Short-term actions include community awareness creation, institutional strengthening, and capacity-building initiatives. Medium- and long-term interventions involve climate-resilient infrastructure development, ecosystem restoration, and the promotion of nature-based and technological solutions. For instance, the installation of water harvesting and storage systems is intended to mitigate drought and saline intrusion, while mangrove rehabilitation and drainage restoration will reduce the impacts of flooding, coastal erosion, and storm surges.

The formulation of these adaptation actions integrates local knowledge with scientifically validated best practices drawn from Kenya's National Adaptation Plan (2015–2030), the National Climate Change Action Plan (2023–2027), and the Kwale County Climate Change Action Plan (2023–2027). Gender equity and social inclusion principles are mainstreamed throughout to ensure that adaptation benefits are distributed equitably. Women, who bear a disproportionate burden during droughts and resource scarcity, are prioritized in livelihood diversification, leadership, and financial empowerment programs. Youth are similarly engaged through technology-based initiatives such as climate information dissemination, ecosystem monitoring, and innovation in sustainable resource use.

Currently, there are ongoing community-led projects in Vanga Ward that complement these adaptation priorities, including mangrove restoration, village-based seaweed farming, tree planting initiatives, and local savings and credit groups (VSLAs) that support climate-resilient livelihoods. These initiatives demonstrate the community's commitment to sustainable adaptation and provide a strong foundation upon which the proposed actions will build.

Implementation of the adaptation measures will occur at multiple levels i.e., household, community, ward, county, and national, under the coordination of the Ward Climate Change Planning Committee (WCCPC). Collaborative engagement among local organizations, BMUs, WRUAs, and relevant government institutions will be essential to ensure coherent, inclusive, and sustainable delivery of the adaptation agenda across Vanga Ward.

5.2 Existing climate-resilient and livelihood-enhancing projects

Vanga Ward currently benefits from a diverse network of community-led initiatives, institutional partners, and development organizations implementing climate-resilient and livelihood-enhancing projects (Table 4). These interventions collectively strengthen local adaptive capacity, promote ecosystem restoration, and support sustainable resource use. Existing strategies focus on forest restoration, water access, blue carbon initiatives, sustainable agriculture, and community-based financial inclusion, demonstrating strong collaboration between the community, civil society, and county agencies.

Key ongoing initiatives include the establishment and maintenance of tree nurseries and reforestation programs led by groups such as Umba Self-Help Group (S.H.G), Mwalewa C Kids Care, NEEMA, Mwalunga Chumbi CBO, Tumaini Care Giver, Nyota Njema Disabled Group, and Gonja Tree Planters, with technical and financial support from Plan International. These efforts contribute to catchment restoration, soil conservation, and increased vegetation cover under the broader county ecosystem restoration strategy.

Seaweed farming, implemented by Jimbo Youth Group, continues to expand as a climate-smart livelihood option for youth and women, offering income diversification while contributing to marine ecosystem health. The Vanga Blue Forest project is implementing both carbon credit initiatives and borehole drilling programs (notably in Kiwegu B), linking coastal restoration with improved freshwater access. In parallel, the Community Development Trust Fund (CDTF) has supported similar water access projects in Jimbo, enhancing drought resilience through community-managed boreholes.

Mangrove conservation and rehabilitation form a cornerstone of ongoing environmental management efforts, spearheaded by groups such as Vumilia Nguvu Kazi (Kiwegu), Jimbo Environmental Group, Mwayugu Mariculture (Vanga), and Mwambiweje Women Group (Vanga). These organizations manage mangrove nurseries, replant degraded sections, and promote community stewardship, aligning with national frameworks such as the National Mangrove Ecosystem Management Plan (2017–2027).

Complementary socio-economic initiatives include table banking programs implemented by COMRED in collaboration with Jimbo BMU, which enhance household financial security and provide capital for small-scale enterprises. World Vision Kenya supports capacity building in agroforestry for carbon credit generation, while the National Agricultural Value Chain Development Project and Kids Care promote kitchen gardening and poultry farming to strengthen food security at household level.

Plan International continues to play a central role in institutional capacity building through training on forestry management, seaweed farming, agricultural trial plots, orchard establishment, and Village Savings and Loan Associations (VSLAs). These initiatives complement ongoing county strategies such as the Kwale County Climate Change Action Plan (2023–2027), County Integrated Development Plan (CIDP 2023–2027), and the Kwale County Forest and Landscape Restoration Strategy, ensuring that community actions are aligned with county and national adaptation frameworks.

Table 4: Existing climate-resilient and livelihood-enhancing projects in Vanga Ward

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
Tree Nursery and Tree Planting	Umba Self Help Group (S.H.G), Mwalewa C Kids Care, NEEMA, Mwalunga Chumbi CBO, Tumaini Care Giver, Nyota Njema Disabled Group, Gonja Tree Planters	Mwalewa C, Ngweneni, Mikoroshoni	Environmental Restoration	Supports afforestation and reforestation under the Kwale County Forest and Landscape Restoration Strategy; enhances vegetation cover and soil stability.
Seaweed Farming	Jimbo Youth Group	Jimbo, Vanga coastline	Livelihood Diversification / Marine Ecosystem Management	Promotes nature-based livelihoods and women/youth economic empowerment aligned with the Blue Economy Strategy.
Carbon Credit Project	Vanga Blue Forest	Vanga, Kiwegu B	Carbon Sequestration / Ecosystem Restoration	Implements blue carbon initiatives and carbon offset systems consistent with the National Mangrove Ecosystem Management Plan (2017–2027).
Mangrove Nursery and Restoration	Vumilia Nguvu Kazi (Kiwegu), Jimbo Environmental Group, Mwayugu Mariculture (Vanga), Mwambiweje Women Group (Vanga)	Vanga, Kiwegu, Jimbo	Coastal Ecosystem Restoration	Focuses on mangrove rehabilitation, shoreline stabilization, and climate regulation; contributes to blue carbon storage.
Table Banking and VSLA Support	COMRED (to Jimbo BMU)	Jimbo	Social Protection / Financial Inclusion	Strengthens household savings and access to credit; enhances adaptive capacity through women-

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
				and youth-led savings groups.
Borehole Drilling and Solarization	Vanga Blue Forest (Kiwegu B), Community Development Trust Fund (CDTF - Jimbo)	Kiwegu B, Jimbo	Water Security / Drought Resilience	Enhances access to clean water and reduces time spent on water collection, aligning with the County Water Investment Plan.
Agroforestry for Carbon Credit	World Vision Kenya	Mwalewa C, Ngweneni	Sustainable Agriculture / Carbon Sequestration	Promotes tree-based farming systems and household-level carbon projects integrated into climate-smart agriculture.
Kitchen Gardening and Poultry Farming	Kids Care, National Agricultural Value Chain Development Project (NAVCDP)	Mwalewa C, Ngweneni, Mikoroshoni	Food Security / Climate-Smart Agriculture	Enhances household nutrition, diversifies income, and promotes efficient water use for small-scale horticulture.
Capacity Building on Forestry, Seaweed Farming, Agriculture Trial Farms, Orchard Farming, and VSLAs	Plan International	Mikoroshoni, Mwalewa C, Ngweneni, Sega D	Institutional Strengthening / Livelihood Diversification	Builds local technical capacity and integrates gender-responsive livelihood models; supports ongoing community adaptation programs.
Community Tree and Mangrove Restoration for Ecosystem Protection (Existing Ward Strategy)	Gonja CFA, WCCPC, Plan International	Mikoroshoni, Kiwegu	Environmental Restoration / Community-Led Adaptation	Aligns with Kwale County Climate Change Action Plan (2023–2027); complements national reforestation targets.

5.3 Adaptation actions

Adaptation actions in Vanga Ward are designed as an integrated and gender-responsive climate resilience framework that protects livelihoods, strengthens ecosystems, and safeguards the well-being of all population groups across the four villages; Mikoroshoni, Mwalewa C, Ngweneni and Sega D (Table 5). These actions respond to six priority climate hazards; drought, floods, pests and diseases, strong winds, environmental degradation and human–wildlife conflicts and are anchored in the Kenya Climate Change Act (2023), the National Adaptation Plan (2015–2030), the Kwale County Climate Change Act (2021), and national frameworks on gender equality, child protection, and sexual and reproductive health and rights (SRHR). Implementation follows a phased approach comprising short-term interventions (≤1 year), medium-term measures (≤3 years), and long-term interventions (>3 years), all anchored in gender equity, social inclusion, SRHR continuity, and community ownership to ensure climate benefits are distributed fairly and sustainably.

Under Thematic Area One, Water Security and Management focuses on strengthening household, institutional and community access to reliable and safe water resources across Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages. In the short term, households will prioritize rainwater harvesting through the installation of gutters and storage tanks at an estimated cost of KES 50,000 per household, reducing the time burden on women and girls who undertake most water collection. Immediate ecosystem rehabilitation will be undertaken through desilting of the community water pan along the Mutsongo–Kimori stretch in Ngweneni Village, budgeted at KES 1,000,000, thereby restoring water storage capacity for domestic and livestock use. These short-term actions are complemented by sensitization of women and youth on household water treatment practices, implemented through barazas facilitated by CHPs, NGOs and KWAWASCO at KES 20,000 per village.

Medium-term measures focus on institutional strengthening through the installation of 10 m^3 high-capacity water tanks and rooftop harvesting systems in one school per village, except in Mwalewa C where a church will serve as the anchoring institution, costed at KES 400,000 per institution. These facilities will support school WASH systems and provide fallback water supplies during drought. The programme will also drill and solarize one borehole per village, each budgeted at KES 5,000,000, to ensure perennial water availability for domestic and small-scale irrigation use. These interventions will be delivered collaboratively through Kwale County Government (FLLoCA), NGOs, private sector partners and KWAWASCO.

Long-term actions advance strategic water supply expansion through the installation of solar-powered water pumps along River Umba, coupled with elevated storage tanks to enhance water reliability for households and women-led livelihood systems. Each solar pumping unit is estimated at KES 70,000. To further enhance resilience, the construction of additional water pans in Mikoroshoni, Mwalewa C and Ngweneni at KES 3,000,000 per water pan will store excess rainfall and provide year-round supply for irrigation and livestock. These long-term interventions collectively aim to reduce climate-related water stress while improving gender-equitable access to safe water.

Thematic Area Two focuses on strengthening household food security and ecosystem productivity through climate-smart agriculture and resilience design from homestead to watershed level. In the short term, women and youth will be capacitated on resilience design techniques, soil fertility enhancement, water retention using zai pits, mulching and efficient farm layout, delivered through county extension services and NGOs at KES 10,000 per farmer per 12-month cycle. Training will also cover climate-smart agriculture, including drought-tolerant crop varieties (cassava, katumani maize, sorghum), hay and silage making, as well as sensitization on index-based crop and livestock insurance.

Medium-term interventions include distributing certified seeds, fertilizers, pesticides and livestock inputs to trained groups in all four villages. Production support is costed at KES 55,000 per acre for maize, KES 40,000 per acre for cassava, and KES 10,000 per livestock farmer. Select women- and youth-led farms will be supported to fully implement resilient design techniques, forming demonstration farms for continuous peer learning. Establishing a Ward-level Farmer Field School (FFS) will institutionalize resilient farming practices and ensure continuous technical support.

Long-term actions prioritize market strengthening by aggregating produce from women and youth to enhance bargaining power and value addition. Livestock diversification will be supported through pasture establishment at KES 55,000 per acre, creation of fodder seed banks, and distribution of Gala goats (KES 15,000 each) and dairy cows (KES 100,000 each). These interventions reduce climate sensitivity while improving livelihood stability.

In addition, the thematic area focuses on restoring degraded ecosystems and enhancing nature-based solutions. In the short term, women, youth and PADs will be trained to establish nurseries for priority tree species for agroforestry, boundary planting and woodlot establishment. Training is budgeted at KES 5,000 per trainee per day, and each nursery will cost KES 75,000 to establish. Training on briquette production will also be provided to reduce dependence on charcoal and mitigate forest degradation.

Medium-term interventions include providing households with 100 priority seedlings per home at KES 5,000 per household, rehabilitating degraded sections of Gonja Forest and Kaya Sega at KES 150,000 per acre, and strengthening CFAs to implement PFMPs. Provision of energy-saving jikos (KES 3,000) and kilns (KES 7,000) will reduce pressure on forest resources, enhancing long-term sustainability.

Long-term actions emphasize regulation and controlled extraction along River Umba, implemented by NEMA, WRA and the County Government. Sensitization barazas at KES 20,000 per village will promote sustainable sand harvesting, riverbank protection and flood mitigation, ensuring ecological integrity of the watershed.

Thematic Area Three strengthens adaptive capacity through diversified income streams and social safety nets. Short-term actions include training women, youth and PADs in alternative livelihoods such as kitchen gardening, beekeeping, poultry production and briquette making, delivered at KES 5,000 per trainee per day. VSLA governance structures will be strengthened to promote financial literacy and enhance adaptive capacity.

Medium-term actions include provision of livelihood inputs, seeds and organic fertilizers, modern beehives (KES 7,000) and egg incubators (KES 100,000), to facilitate enterprise establishment. VSLAs will be supported to invest excess savings in green financial products such as Safaricom's Green Bond.

Long-term actions include constructing a community social hall in Mwalewa C, guided by BOQs, to serve as a multifunctional income-generating centre supporting women and youth enterprises.

Thematic Area Four addresses structural vulnerabilities to floods, strong winds and drought. Short-term actions include training community emergency response teams across all four villages at KES 5,000 per trainee, alongside tree planting in Sega D to serve as a windbreak system. These foundational measures strengthen preparedness and immediate response to climate hazards.

Medium-term interventions include a food-for-work programme, previously successful in Kwale County, to engage youth in restoring drainage channels, cleaning culverts, constructing gabions and undertaking tree planting for erosion control. Early Warning Systems (EWS) will be developed and operationalized in all villages at KES 70,000 per village, improving access to predictive information on floods, droughts, and pest outbreaks. Community emergency response teams will be equipped with PPEs and first-aid kits, and household-level pit latrines will be constructed in Ngweneni at KES 100,000 per unit to enhance sanitation and prevent disease outbreaks during floods.

Long-term actions involve constructing raised footbridges and culverts at flood-prone crossing points in Mwalewa C, Ngweneni and Sega D, and improving road drainage systems through the installation of new culverts to channel stormwater safely. All long-term infrastructural works will be guided by BOQs and implemented in partnership with Kwale County Government, KERRA, KMD, NDMA, Kenya Red Cross and NGOs.

Thematic Area Five addresses rising human–wildlife conflicts (HWC) especially in Mwalewa C and villages near Gonja Forest. Short-term actions include community sensitization on conflict hotspots, safe behaviour to reduce encounters, and reporting mechanisms to KWS through the emergency hotline. Each sensitization forum will cost KES 20,000 per village. Long-term strategies involve strengthening coordination between KWS, county authorities and communities to institutionalize conflict mitigation, improve surveillance and enhance access to compensation pathways.

Thematic Area Six centers on integrating gender equality, youth empowerment and inclusion of PADs into climate adaptation. In the short term, targeted GBV anticipatory actions will be developed for drought, floods and strong winds, ensuring safe access to services for women and girls. Existing women's associations such as VSLAs will be used to disseminate information on rights, helplines and GBV response systems at KES 30,000 per VSLA. Youth boot camps addressing drug abuse, reproductive health and life skills will be implemented at KES 100,000 per camp, and children and youth will be vaccinated at local health facilities to reduce outbreak risks during climate shocks.

Medium-term actions include assessing and refining Early Warning Systems for climate-linked GBV anticipatory responses, and strengthening inclusive governance through training women, youth and PADs at KES 5,000 per person per day. GBV response mechanisms, including transfer vehicles and specialized first-aid and rape kits, will be procured based on BOQs. School feeding and scholarship support will be anchored on needs assessments to safeguard climate-vulnerable learners.

Long-term interventions include developing and implementing a comprehensive Ward GBV anticipatory action plan (costed at KES 1,000,000 for consultancy services), constructing a county-level drug rehabilitation centre, and facilitating PADs to access mobility devices, assistive technology and entrepreneurial training through stakeholder engagement forums budgeted at KES 50,000.

Table 5: Vanga Ward Adaptation Actions

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
Thematic area 1: Water Security and Management				
Rainwater harvesting through Installation of gutters on house roofs and water tanks at household level in Mikoroshoni, Mwalewa C, Ngweneni, and Sega D Villages Desilting community water pan along Mutsongo–Kimori in Ngweneni Village	Rainwater harvesting through installation of gutters and high-capacity water tanks (10m^3) in schools– one school in each village to enhance water access by women and children Mwalewa C installation to be done in one of the churches due to lack of a school	Drilling and solarizing boreholes in Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages- one in each village	Kwale County Government-FLLoCA, NGOs and Private Sector	50,000/HH 400,000/School 5,000,000/borehole 1,000,000/desilting water pan
	Strategic (where embankment for water intake is viable) installation of solar powered water pumps along River Umba and elevated storage tanks to provide domestic and irrigation water sources for women and girls in Mikoroshoni, Mwalewa C, Ngweneni, and Sega D villages	Construction of a water pan in Mikoroshoni, Mwalewa C and Ngweneni villages		70,000/solar water pump 3,000,000/water pan
Sensitizing women and youth on water treatment in Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages			KWAWASCO, Kwale County Government, CHPs, NGOs and Private Sector	20,000 village baraza
Thematic area 2: Food security, environmental restoration and livelihoods				
Training and supporting women and youth on Resilience Design techniques from homestead to watershed level (Soil fertility improvement, water conservation/retention (e.g. Zai Pits), farm design to maximize on time and use of local inputs to reduce farming costs).	Supporting select women and youth lead farmers to fully implement resilient design techniques including climate smart agriculture to serve as demonstration farms for continuing learning by other farmers to enhance sustainability of the interventions	Establishing a Farmer Field School in the Ward	Kwale County Government, KARLO, NGOs and Private Sector	Farmer Field Schools (FFFs) 10,000 per farmer for a 12-month cycle
Train women and youth on Climate Smart Agriculture (CSA) (drought tolerant crops such as cassava, katumani maize and sorghum), silage/hay making to address livestock	Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and extension services across Mikoroshoni,	Aggregating women and youth farmers' produce to enhance farm produce marketability and enhance value		Production costs: Maize 55,000 per acre and Cassava 40,000 per acre

feed challenges during drought in Mikoroshoni, Mwalewa C, Ngweneni, and Sega D villages	Mwalewa C, Ngweneni, and Sega D Villages to trained women and youth groups	addition and agribusiness opportunities		10,000 for livestock farmers per year or crop cycle (Free if it is provided by the government)
Sensitize and facilitate access to information on crop and livestock insurance (index-based drought coverage) targeting women and youth groups involved in CSA	Establish livestock fodder seed banks, pastures, and introduce drought-tolerant livestock breeds e.g., Gala goats/Dairy cows for women and youth groups in Mikoroshoni, Mwalewa C, Ngweneni villages			Village barazas at 20,000, 55,000/acre for pasture, 15,000 /Gala goat, 100,000 per dairy cow
Facilitate/train women and youth to establish tree nursery of priority species (Annex 1) for on-farm tree production (Agroforestry, woodlots and boundary planting) in Mikoroshoni Mwalewa C, Ngweneni and Sega D villages	Provision of priority tree and fruit seedlings at household level in Mikoroshoni Mwalewa C, Ngweneni and Sega D Villages Rehabilitate degraded forests and landscapes (Gonja forest and Kaya Sega) Support Gonja Forest CFA to implement PFMP		KFS, KEFRI, Kwale County Government, CFAs, NGOs, Private Sector	5,000/day per trainee (40 pax from the four villages), 5,000/HH (100 seedlings/HH), 75,000/tree nursery, 150,000/acre for rehabilitation, CFA support based on PFMP budget lines and priorities
Training women, youth and PADs on making organic briquettes across Mikoroshoni Mwalewa C, Ngweneni and Sega D Villages	Provision of energy saving jikos and kiln across Mikoroshoni Mwalewa C, Ngweneni and Sega D Villages		Kwale County Government, NGOs, Private Sector	5,000/day per trainee (40 pax from the four villages) 3,000/energy saving jiko & 7,000/kiln
Sensitization on riverbanks management and flood control and sand harvesting impacts along River Umba	Regulation of sand harvesting along River Umba		NEMA, WRA, Kwale County Government, NGOs, Private Sector	20,000/village baraza
Thematic area 3: Diversified Livelihoods and Social Protection				
Training women, youth and PADs on alternative livelihoods such as kitchen gardening (horticultural crops),	Provision of alternative livelihoods establishment materials/Equipment; kitchen gardening (seeds and organic	Construction of a community social hall in	Kwale County Government, NGOs, Private Sector	5,000 per person per day for training

beekeeping, poultry farming and briquettes making in Mikoroshoni Mwalewa C, Ngweneni and Sega D villages	fertilizers) modern beehives and incubators in Mikoroshoni, Mwalewa C, Ngweneni and Sega D Villages	Mwalewa C village to generate income		7,000/modern beehive & 100,000/egg incubator BOQ required for social hall construction
Train VSLAs on Governance and alternative regenerative investment opportunities such as green bonds and money market funds to diversify interest income from surplus savings	Promote VSLAs excess savings investments in reputable interest earning schemes such as the green bond by Safaricom			100,000 per VSLA
Thematic area 4: Community infrastructure and disaster management				
Train selected community emergency response teams in Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages Undertake tree planting to address strong winds in Sega D	Design and implement a food for work programme during drought (this has been done before in the county) targeting the youth to restore drainage channels, clean culverts, construct gabions and plant trees for erosion and flood control	Construct raised footbridges/culverts at flood-prone crossing points in Mwalewa C, Ngweneni and Sega D villages Improve road drainage and install culverts to channel floodwaters in Mwalewa C	Kwale County Government, KERRA, NGOs and Private Sector	5,000/trainee BOQs required based on numbers and reputable training institutions
	Develop and operationalize Early Warning Systems (EWS) for floods, drought, pests and diseases (forecasts via chiefs, barazas, media) in Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages	Equip community emergency response teams (PPEs, first aid kits) in Mikoroshoni, Mwalewa C, Ngweneni and Sega D villages Construction of pit latrines at household level in Ngweneni Village	KMD, NDMA Kwale County Government, Ministry of Interior and National Administration Kenya Red Cross, NGOs and Private Sector	70,000/village for EWS 100,000/pit latrine
Thematic area 5: Conflicts resolution (Human Wildlife Conflicts)				
Sensitize Mwalewa C and Gonja forest adjacent communities on potential conflict hotspots, minimizing human-			KWS, Kwale County Government, NGOs, Private Sector	20,000/village baraza

wildlife interactions and reporting to KWS (hotline) as well as compensation				
Thematic area 6: Gender and social inclusion to enhance adaptation action				
Design targeted GBV anticipatory actions for drought, floods and strong winds climate hazards across Mikoroshoni Mwalewa C, Ngweneni and Sega D villages	Develop and implement a Ward GBV anticipatory action plan in collaboration with relevant partners		National Government, Kwale County Government and NGOs, National Police Service, NGOs, GVRC, COVAW, HAK	1,000,000 for consultancy services
Assess the existing early warning systems to produce and transmit reliable and timely information and forecasts for the implementation of GBV anticipatory actions ahead of climate hazards	Strengthen gender-responsive governance- inclusion of women, youth and PADs through capacity building across Mikoroshoni Mwalewa C, Ngweneni and Sega D villages		Kwale County Government and NGOs	5,000/person per day for training
Use of existing women associations such as VSLAs to train women and girls on their rights and GBV response strategies, helplines and recovery centers for women and girls as well as the role of men in peaceful conflict resolution	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		Kwale County Government, Civil Society, NGOs, Private Sector	30,000/VSLA BOQ for GBV response mechanisms
Organize and hold youth boot camps on guidance and counselling focusing on drugs, antisocial behavior and reproductive health	Promote vaccination of children and youth against diseases such as cholera which could arise and spread on account of climate hazards Support school feeding programmes and scholarships	Construct drug rehabilitation centre targeting the youth (this can be done at the county level)	National Government, NACADA, Kwale County Government, NGOs, Private Sector	100,000/bootcamp Vaccination at local health facility BOQ required for drug rehabilitation centre School feeding and scholarships needs assessment
Facilitate PADs to link with organizations and companies that can provide infrastructural assistance such			Kwale County Government, NGOs, Private Sector	50,000 for meetings

as wheelchairs, specialized training and entrepreneur skills				

5.4 Resource Mobilisation

Implementation of the Ward Adaptation Plan will rely on a diversified mix of financial, human, and technical resources drawn from county, national, donor, and community-based funding streams. The primary sources include county budget allocations, where relevant departments such as Water, Agriculture, and Infrastructure will integrate adaptation priorities into their Annual Development Plans and sector budgets, supported by at least KES 2–3 million annually earmarked for ward-level climate resilience as required under the Climate Change Act, 2023. Additional financing will be accessed through the Kwale County Climate Change Fund (CCCF) established under the County Climate Change Act, 2022, which channels grants from development partners such as the World Bank's FLLoCA programme to locally led adaptation initiatives. At the national level, Vanga Ward will benefit from the National Drought Emergency Fund and other climate financing mechanisms such as the Kenya Adaptation Fund Programme, coordinated through the National Treasury's Climate Finance Unit. Development partners and NGOs, including Plan International, WWF, World Vision, and COMRED, will continue to support ecosystem restoration, capacity building, and livelihood diversification projects, while community contributions through in-kind labour, savings groups, and user fees will promote ownership and sustainability. The private sector will be engaged through partnerships for water infrastructure, crop and livestock insurance, and digital early warning systems, complemented by corporate social responsibility initiatives. Overall, the ward expects to mobilize approximately KES 20–30 million over five years from this blended financing framework, ensuring continuity and long-term sustainability of adaptation interventions.

6 Chapter 6: Adaptation Monitoring and Evaluation

6.1 Introduction

Monitoring, Evaluation, and Learning (MEL) are critical components of the Ward Adaptation Plan to ensure that the proposed actions are achieving their intended outcomes and to facilitate continuous improvement. This chapter outlines how Vanga Ward will track progress of adaptation interventions (Monitoring), assess their effectiveness and impact (Evaluation), and capture lessons to inform future climate-resilience planning (Learning). A robust M&E framework not only provides accountability to the community and funders but also enables adaptive management, meaning the Ward Climate Committee can adjust strategies based on what is working or not working on the ground. Given the long-term nature of climate change, M&E will help maintain momentum and demonstrate how resilience is being built incrementally. The approach will utilize participatory methods so that community members are involved in monitoring their own resilience, alongside technical indicators aligned with county and national reporting requirements. The Climate Change Act 2023 reinforces the importance of reporting and transparency in climate actions, and this plan's M&E system will align with those national guidelines while tailoring to local context.

6.2 The Adaptation Theory of Change

The theory of change is a conceptual model that explains how the planned interventions will lead to the desired climate resilience outcomes. In simplified form:

- **Inputs/Activities:** Financial resources, training sessions, infrastructure built, seedlings planted, policies enforced (these are the actions from Chapter 4).

- **Outputs:** The immediate results of those activities – e.g., number of boreholes drilled and operational, farmers trained in climate-smart practices, trees planted and surviving, households with new water tanks, functional ward climate committee meetings held, etc.

- **Outcomes:** The short to medium-term changes due to those outputs – e.g., increased water access during dry season, adoption of drought-tolerant crops on farms, reduced time spent fetching water (especially for women), increased savings in VSLAs, decreased crop losses from pests, improved condition of forests.

- **Impact (Goal):** The higher-level objective is enhanced climate resilience and adaptive capacity of the Vanga Ward community. This means the community can better withstand droughts, floods, and other hazards without suffering devastating losses, and can recover more quickly. It also includes improved livelihoods and food security under a changing climate, and a reduction in climate-induced emergencies over time.

The theory of change for Vanga can be summarized as: "IF we provide reliable water sources, promote resilient farming and livelihoods, strengthen natural resource management, and enhance local institutions' capacity, THEN the community's exposure and sensitivity to climate hazards will decrease and their adaptive capacity will increase, BECAUSE they will have the necessary resources, knowledge, and organization to cope with climate stresses." This will lead to the ultimate impact of a more climate-resilient Vanga Ward with secure livelihoods and wellbeing despite climate change.

This logical flow will guide M&E. Each level (input -> output -> outcome -> impact) will have specific indicators to check. For instance, training farmers (input) should lead to output of farmers implementing new techniques, which yields outcome of yield stability in drought, contributing to impact of overall food security. If a link is failing (e.g., outputs achieved but

expected outcome not observed), it signals a need to investigate and possibly modify the strategy. Importantly, this plan assumes certain enabling conditions (or intermediate outcomes) as part of the theory of change: community engagement remains strong, the local government continues supportive policies, and external shocks beyond climate (like economic downturns or conflicts) do not severely undermine efforts. M&E will also keep an eye on these assumptions. In the theory of change, gender and social inclusion are cross-cutting, by involving women and youth in decision-making (output), we expect more equitable resilience building (outcome), contributing to a more cohesive and adaptable society (impact). Likewise, capacity building of institutions (output) yields better coordination and resource access (outcome), leading to sustainable implementation (impact). By explicitly stating this theory, all stakeholders have a common understanding of what success looks like and how we plan to get there, which will inform what we monitor and evaluate.

6.3 Performance Indicators for Adaptation

Based on the theory of change, a set of Performance Indicators will be used to measure progress and effectiveness. These indicators will be both quantitative and qualitative, and they align with national adaptation M&E frameworks where possible (Kenya's NAP/NCCAP indicators) as well as community-defined measures of success. The performance indicators are structured at various levels:

Input/Output Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Number of household rainwater harvesting systems installed (gutters + tanks).
   - Number of schools equipped with high-capacity rainwater harvesting systems.
   - Number of water pans constructed or desilted.
   - Length/coverage of pipelines and kiosks constructed (e.g., Mkwiro–Wasini).
   - Number of manual and solar-powered pumps installed on existing wells.
   - Number of women, youth, and PADs using solarised/manual hand pumps.
   - Number and percentage of women, youth, and PADs in water management committees.
   - Number of village water storage structures constructed (Majibao, elevated tanks).
   - Amount of budget allocated and utilised for water infrastructure.
   - Number of boreholes drilled, equipped, and solarised.
   - Number of boreholes fully operational and supplying water year-round.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS
   - Number of farmers trained on Resilience Design techniques.
   - Number of farmers trained on CSA practices (land prep, planting, pest control, post-harvest).
   - Acreage under drought-tolerant crops supported with certified seeds.
   - Number of Farmer Field Schools established and functional.
   - Number of demonstration farms established by women and youth lead farmers.
   - Number of farmers receiving inputs (certified seeds, fertilizers, pesticides).
   - Number of households supported with organic manure production and composting.
   - Number of farmers adopting terracing, zai pits, swales, and other soil/water conservation measures.
   - Number of livestock fodder seed banks and pasture plots established.
   - Number of households receiving improved livestock breeds (Gala goats, dairy cows).
   - Number of farmers sensitised on crop/livestock insurance.
   - Number of households supported with grain storage equipment (hermetic bags, grain banks).

- Number of village pest and disease surveillance teams established and trained.
- Number of households receiving indigenous/fruit seedlings.
- Number of community seedling distribution events held.
- Number of on-farm tree planting sessions conducted.
- Number of mangrove nurseries supported and maintained.
- Number of acres of mangrove restored through enrichment planting.
- Number of community trainings on mangrove protection and sustainable harvesting.
- Number of NBS windbreaks (Neem, Casuarina) planted across all villages.
- Number of environmental conservation awareness sessions held.
- Number of households adopting sustainable charcoal alternatives or clean energy.
- Budget allocated and utilised for restoration activities.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION
   - Number of women and youth trained in alternative livelihoods (eco-tourism, beekeeping, poultry).
   - Number of modern beehives, incubators, or eco-tourism boats distributed.
   - Number of households supported with livelihood start-up kits.
   - Number of community groups formed for value addition and market linkages.
   - Number of training sessions on business development, value addition, and financial literacy.
   - Number of social protection initiatives implemented (food distribution, cash transfer mobilizations).
   - Number of school feeding programmes supported.
   - Budget allocations for livelihood diversification implemented.

4. COMMUNITY INFRASTRUCTURE AND HUMAN SETTLEMENTS
   - Number of culverts reconstructed or improved with proper drainage.
   - Number of flood-prone crossing points fitted with raised footbridges.
   - Number of youth mobilised in food-for-work programmes for drainage, gabions, and erosion control.
   - Number of villages with functional flood Early Warning Systems (EWS) linked to KMD.
   - Number of community dissemination channels established (chief barazas, noticeboards, SMS).
   - Number of community emergency response teams trained.
   - Number of community emergency response teams equipped (boats, PPE, first-aid kits).
   - Number of designated disaster assembly points established and labelled.
   - Number of community trainings on resilient building practices conducted.
   - Number of households implementing safer roofing techniques (replacing strings with nails).
   - Budget allocated and utilised for infrastructure upgrades (roads, culverts, drainage).

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)
   - Number of community sensitization sessions on human–wildlife conflict hotspots.
   - Number of households supported with HWC reporting mechanisms (hotline awareness).
   - Number of kilometres of solar-powered fencing installed in hotspot areas (e.g., Kiwambale).
   - Number of community trainings on safe behaviour in crocodile, baboon, and snake-risk zones.
   - Number of reported HWC cases logged and submitted to KWS for follow-up.
   - Funding allocated and utilised for HWC mitigation (fencing, awareness, surveillance).

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS
   - Number of gender-responsive governance trainings conducted.
   - Number and percentage of women, youth, and PADs participating in ward committees.
   - Number of VSLAs supported with GBV awareness and rights training.
   - Number of GBV anticipatory action sessions conducted for drought, floods, and strong winds.
   - Number of GBV response kits (first-aid kits, rape kits) distributed to health facilities.
   - Number of early warning systems integrating GBV alert messaging.
   - Number of PADs supported with assistive devices, rehabilitation linkages, or livelihood equipment.
   - Number of youth bootcamps conducted on drugs, behaviour change, and reproductive health.
   - Number of awareness campaigns on GBV helplines and safe reporting mechanisms.
   - Number of meetings with PAD support organizations facilitated.

Outcome Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Reduction in average time spent collecting water (minutes/day), disaggregated by women, youth, and PADs.
   - Increase in household water storage capacity (litres/HH).
   - Percentage of water facilities functional year-round.
   - Increase in number/percentage of women, youth, and PADs with independent access to water sources (boreholes, solarised pumps, kiosks).
   - Increased reliability of water supply during drought periods (days/months of service continuity).

2. FOOD SECURITY, ENVIRONMENTAL RESTORATION AND LIVELIHOODS
   - Stability of crop yields over 3–5 seasons among supported farmers.
   - Improvement in soil health indicators, including:
   - Increase in soil organic matter (%)
   - Increase in soil moisture retention capacity
   - Improvement in soil nutrient status (N, P, K)
   - Improved aggregate stability and soil structure
   - Percentage of farmers adopting integrated soil and water conservation (terracing, zai pits, swales).
   - Increase in acreage under diversified and climate-resilient farm systems.
   - Increase in household-level adoption of post-harvest management technologies (grain banks, hermetic bags).
   - Percentage of women, youth, and PADs participating in CSA and Resilience Design training cycles.
   - Increase in on-farm and community tree cover (%) over baseline.
   - Increase in mangrove seedling survival rates and total acreage restored.
   - Reduction in incidents of illegal harvesting, charcoal burning, and mangrove cutting.
   - Increase in households adopting windbreak and shelterbelt systems.
   - Increased community participation in environmental conservation and sustainable harvesting practices.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION
   - Percentage increase in household income from diversified livelihoods (eco-tourism, poultry, beekeeping, seaweed value addition).
   - Increase in the number of households with multiple income streams across seasons.
   - Percentage of women, youth, and PADs accessing new livelihood opportunities.
   - Increased financial literacy and business skills among trained groups.
   - Increase in households benefiting from school feeding, cash transfers, or food-for-work

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT
   - Reduction in households exposed to flood-prone or erosion hotspots.
   - Percentage of villages receiving timely Early Warning System (EWS) alerts.
   - Community response rate to EWS alerts (attendance at assembly points, evacuation compliance).
   - Reduction in flood-related damages to dwellings and access roads.
   - Level of preparedness and readiness of community emergency response teams (measured via drills).
   - Increased adoption of wind-resilient building practices in households.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)
   - Reduction in frequency of human–wildlife conflict incidents (crocodile attacks, baboon raids).
   - Increase in the number of households protected by solar-powered fences.
   - Increase in community capacity to report and respond to HWC events (via KWS hotline).
   - Increased community compliance with safe-behaviour protocols in HWC hotspots.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS
   - Increase in the number and proportion of women, youth, and PADs participating in ward-level governance and adaptation committees.
   - Reduction in drudgery for women, youth, and PADs due to improved water access and resilient infrastructure.
   - Percentage increase in women's and PADs' access to credit, extension services, and productive resources.
   - Increase in women's ownership of productive assets (livestock, seaweed farms, equipment).
   - Improvement in community knowledge and awareness of GBV risk mitigation during climate hazards.
   - Increased utilisation of GBV response services (safe shelters, first-aid kits, referral pathways).

Impact (Goal) Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Percentage of the ward population with year-round access to safe and reliable water.
   - Increase in average litres of water available per person per day across villages.
   - Long-term reduction in water scarcity–related vulnerabilities for women, youth, and PADs.
   - Sustained decrease in waterborne diseases due to improved water quality and availability.
   - Community-wide resilience to prolonged drought through diversified water sources and storage systems.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS
   - Sustained increase in crop yields above baseline across 3–5 year periods.
   - Long-term reduction in seasonal yield fluctuations linked to droughts, floods, and pest outbreaks.
   - Measurable improvements in soil health at landscape level, including:
   - Higher soil organic carbon (SOC)
   - Increased vegetation cover (NDVI index)
   - Reduced erosion and sediment loss
   - Enhanced household food security across all seasons.
   - Increased biodiversity within integrated farming systems (trees, fodder, pollinator presence).
   - Increases in forest cover, on-farm trees, and mangrove extent compared to baseline.
   - Restoration of degraded landscapes evidenced by improvements in NDVI and tree density.
   - Reduction in illegal harvesting, mangrove cutting, and charcoal burning.
   - Enhanced ecosystem services, including shoreline protection, soil stability, and microclimate regulation.
   - Long-term improvement in habitat quality for biodiversity and fisheries productivity.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION
   - Sustained increases in household incomes from diversified livelihoods (beekeeping, eco-tourism, seaweed, poultry).
   - Improved household resilience to income shocks from climate hazards.
   - Decreased poverty levels due to stable and multi-stream livelihoods.
   - Increased school attendance and nutritional outcomes for children due to social protection schemes.
   - Greater financial stability among women, youth, and PAD households.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT
   - Significant reduction in climate-related damage to homes, roads, and community facilities.
   - Communities demonstrating effective disaster preparedness and response during floods and storms.
   - Long-term reduction in displacement, injuries, and asset losses during climate hazards.
   - Increased structural safety and durability of settlements in strong-wind and flood-prone areas.
5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)
   - Long-term reduction in human injuries, livestock losses, and crop destruction from wildlife.
   - Improved coexistence between communities and wildlife due to effective mitigation (fencing, reporting systems).
   - Reduced economic losses from HWC-related disruptions.
   - Increased safety and mobility for women, children, and PADs in high-risk zones.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS
   - Percentage increase in women's, youth's, and PADs' participation in climate governance and local decision-making structures.

- Strengthened continuity of SRHR, maternal health, and GBV response services during climate hazards.
- Increased leadership roles held by women, youth, and PADs in adaptation implementation.
- Growth in the number of marginalised and vulnerable groups leading or influencing climate adaptation actions.
- Reduction in GBV risks during droughts, floods, and strong winds due to anticipatory action systems.

For each indicator, we will establish baselines (where we are starting from, using historical data or initial surveys) and set targets for 1-year, 3-year, 5-year intervals. For example: baseline of 0 operational boreholes, target 3 by 2025; baseline average maize yield 8 bags/acre in normal year and 2 in drought year, target to achieve 5 in a drought year with adaptation by 2027; baseline women fetching water 5 km, target <1 km by 2024, etc.

Data for these indicators will come from various sources: routine project monitoring (engineer reports for infrastructure, training attendance sheets), surveys and focus groups for outcomes (e.g., seasonal household surveys to gauge water access, yields, or time saved), and existing data (health clinic records for disease incidence, meteorological records for hazard frequency). All indicators will be disaggregated by gender where applicable to monitor differential benefits. Also, where feasible, by village to see if some areas lag behind. Measuring adaptation success is complex because it's about avoided losses and capacity built, but these indicators together provide a robust picture. The Ward Committee will use a simple M&E matrix listing each indicator, baseline, target, data source, frequency of data collection, and responsible person.

6.4 Adaptation Reporting and Learning

Reporting: Vanga Ward will establish a clear reporting system to ensure information flows to all stakeholders i.e., community members, county authorities, and funding partners. Internally, the WCCPC will prepare quarterly progress reports that document activities done, outputs achieved, challenges encountered, and expenditures made. These will be shared in ward public barazas for transparency (so the community hears what's been done and can give feedback) and submitted to the County Climate Change Unit. Annually, a more comprehensive Ward Adaptation Report will be produced, summarizing the year's progress against the plan's targets and indicators. This annual report can feed into the county's climate change reporting which is required by the Climate Change Act, ensuring Vanga Ward's experiences contribute to county and national adaptation tracking.

The annual report will highlight case studies or success stories (e.g., "Farmer groups in Sega D doubled their harvest despite the 2025 drought by using new techniques a success attributed to our adaptation training"). It will also not shy away from discussing failures or shortfalls, as these provide learning.

At community level, participatory monitoring meetings will be held. For instance, after each rainy season, community members can gather to discuss: did the drainage improvements prevent flooding? Did the new seeds perform better in this season? Such qualitative feedback will be recorded and included in reports.

The Ward Administrator and WCCPC Chair will also report in county forums (e.g., County Adaptation Planning Committee meetings) on progress, championing their ward and possibly leveraging more support by showcasing results.

Learning and Reflective Review: Adaptation is a long journey with uncertainty, so fostering a culture of learning is vital. Twice during the implementation period (e.g., mid-term in year 3, and end-term in year 5), the ward will conduct an outcome evaluation and learning workshop. This involves bringing together community representatives, implementing partners, and possibly external evaluators to systematically review what outcomes have been achieved and why. Key questions: Which interventions are most effective? Are there unintended impacts? What obstacles are reducing effectiveness (maybe maintenance issues, or cultural barriers to adoption)? For example, maybe the uptake of an improved cookstove program is low – through reflection the group might learn women prefer a different design, or fuel is an issue, leading to adjusting that action. These learning workshops will use tools like community scorecards, ranking of interventions by community benefit, and problem tree analysis for any persistent challenges (similar to the PCVA style but now focusing on project performance). Lessons learned will be documented and used to refine the adaptation plan. The Ward Climate Committee is empowered to tweak or reprioritize actions based on evidence. For instance, if a particular crop variety fails, they will note it and try an alternative next season.

Knowledge Sharing: Vanga Ward will also engage in horizontal learning i.e., exchanging experiences with other wards or counties. If something worked really well here (say the VSLA approach to fund water tanks), we can host visitors from nearby wards to showcase it, and similarly we can learn from others (maybe a neighboring ward piloted a solar irrigation scheme that we could replicate). The county climate change unit can facilitate such inter-ward learning exchange visits or meetings.

Feedback to Policy: Through reporting channels, the ward's experiences may also inform higher level policies. For example, if the ward finds that insurance uptake is low because of certain policy constraints, this feedback can go to the national level via NDMA or the National Climate Change Council to refine such programs. The 2023 Act calls for progressive improvement of climate response , our ward's data and stories become part of that national learning loop.

Sustainability of M&E: The Ward Committee will assign an M&E focal person (or sub-committee) to drive these activities. But monitoring won't be just "external evaluation"; it will be integrated into community life e.g., water committees will track water usage and downtime of boreholes as part of their routine, farmer groups will collectively observe and record crop performance. By embedding monitoring tasks into existing structures, we ensure continuity even if specific projects end.

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9 Annexure

1. Priority tree and fruit species

Scientific Name	Common Name
Azadirachta indica	Neem
Afzelia quanzensis	Pod Mahogany / Mkongo
Casuarina equisetifolia	Whistling Pine / Casuarina
Milicia excelsa	African Teak / Iroko / Mvule
Gmelina arborea	Gmelina / White Teak
Tectona grandis	Teak
Grevillea robusta	Silk Oak / Grevillea
Senna spp.	Senna / Cassia
Moringa oleifera	Moringa / Drumstick Tree
Azalea spp.	Azalea
Citrus sinensis	Sweet Orange
Anacardium occidentale	Cashew
Mangifera indica	Mango
Cocos nucifera	Coconut
Carica papaya	Papaya